A STUDY OF RANDOM MATRICES

Sample Covariance Matrices and Spectral Clustering

AN INDEPENDENT STUDY REPORT

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Abstract

[TODO] *Write the abstract once the body is finalized. Suggested skeleton: (1) Open with a one-sentence framing of random matrix theory and why sample covariance matrices matter in high-dimensional statistics. (2) State the scope: we focus on Gaussian sample covariance matrices in Johnstone's spiked model. (3) Outline the journey: foundational probability and linear algebra, sample covariance matrices and the Wishart distribution, low-dimensional fluctuation behavior, the high-dimensional Marchenko-Pastur and Tracy-Widom laws, the spiked covariance model and the BBP phase transition, eigenvector alignment beyond the threshold, and a sketch of the connection to spectral clustering. (4) Note the role of Monte Carlo simulations in illustrating each phase of the argument. (5) Close with the takeaway: the contrast between strong and weak spikes produces a clean phase transition in both eigenvalue location and eigenvector informativeness, with implications for spectral clustering. Avoid em dashes; use clean academic prose.*

Acknowledgements

[TODO] *Acknowledgements paragraph. Suggested mentions: Professor Gustavo Didier (advisor) for the weekly meetings, the reading recommendations (Yao, Zheng, and Bai 2015; Féral and Péché 2009), and the patient guidance through the spiked covariance literature. The second reader (TBD) for serving on the committee. Optional: family, friends, and the Tulane Department of Mathematics.*

Contents

1 Introduction

Why random matrices?

Modern data sets are large and high-dimensional. A single-cell genomics experiment may measure tens of thousands of gene expression levels in a few hundred cells. A finance application may track hundreds of asset returns over a few years of daily data. A computer-vision system may compare millions of image patches in a feature space of several hundred dimensions. In each case the number of variables p is comparable to, or larger than, the number of observations n. The most basic question one can ask of such data is how its variance is distributed across orthogonal directions, and which directions explain the most variation. Answering this question is the job of *principal components analysis*, and the algorithm reduces to computing the eigenvalues and eigenvectors of the *sample covariance matrix*

$$\widehat{\Sigma}_n \;=\; \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{X}_i^\top, \tag{1}$$

where the $\boldsymbol{X}_i \in \mathbb{R}^p$ are the n observed data points, assumed centered.

Classical multivariate statistics tells us what to expect when p is small and n is large. In this low-dimensional regime, $\widehat{\Sigma}_n$ is an accurate estimate of the population covariance Σ, its eigenvalues converge to the population eigenvalues, and its eigenvectors converge to the population eigenvectors. Methods that depend on this convergence, from PCA to factor analysis to the *spectral clustering* algorithms that we will reach at the end of this report, then stand on firm ground.

The classical picture breaks down when p and n are comparable. The sample covariance matrix is no longer a reliable estimate of Σ: its eigenvalues spread into a non-trivial band of random values, and its eigenvectors may carry little or no information about the population directions. Figure 1 makes the contrast concrete. The same Gaussian sampling model produces eigenvalues that concentrate at the population value 1 when $p = 10$ and $n = 1000$, and eigenvalues that spread across a wide interval, with a well-defined limiting density, when $p = 500$ and $n = 1000$.

The mathematical framework that explains and predicts this phenomenon is *random matrix theory*, originally developed for problems in nuclear physics in the 1950s and now a central tool in high-dimensional statistics, signal processing, and machine learning (Yao et al., 2015; Vershynin, 2018; Couillet and Liao, 2022).

What this report does, and how

This report develops, from first principles, the spectral analysis of sample covariance matrices in both regimes, and uses what we learn to understand when methods like spectral clustering can recover hidden structure in high-dimensional data. The development is built bottom-up: every claim is either proved in the text, sketched with a pointer to a full proof, or cited with the cleanest reference we could find. Every theoretical claim is then illustrated with a Monte Carlo simulation in Python; the notebooks live in the `notebooks/` directory of

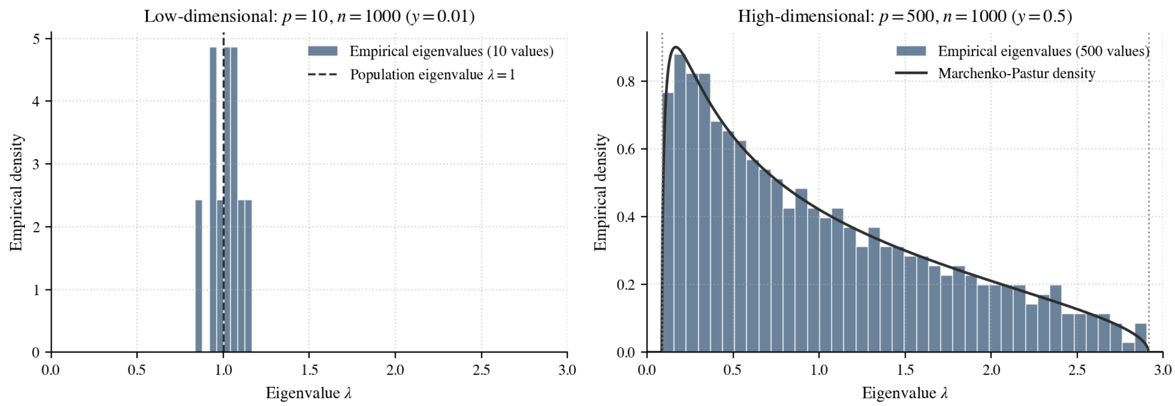

Figure 1: Eigenvalues of the sample covariance matrix $\widehat{\Sigma}_n$ when $\Sigma = I_p$ and the data are Gaussian, for two aspect ratios. *Left:* when $p = 10$ and $n = 1000$ (so $p/n = 0.01$), the ten sample eigenvalues concentrate tightly around the population value $\lambda = 1$, confirming the classical low-dimensional intuition. *Right:* when $p = 500$ and $n = 1000$ (so $p/n = 0.5$), the five hundred sample eigenvalues spread across the interval $[(1-\sqrt{y})^2, (1+\sqrt{y})^2] \approx [0.086, 2.914]$. The empirical density agrees with the Marchenko-Pastur prediction (solid curve). The phenomenon that motivates this report is already visible: in high dimensions, sample eigenvalues are biased, and the spread is a real feature of the geometry, not measurement noise.

the accompanying repository, and the figures they produce are reproduced alongside the corresponding statements.

The starting point is the random variable, the basic building block of probability theory. From there we build to moments and moment generating functions, characteristic functions, modes of convergence, the central limit theorem, random vectors with multivariate normal distributions, and the spectral theorem for symmetric matrices. Each of these objects feeds directly into the analysis of $\widehat{\Sigma}_n$ that follows.

The story arc

The mathematical journey unfolds in three movements.

1. *The low-dimensional regime* (p fixed, $n \to \infty$). The law of large numbers gives entrywise convergence of $\widehat{\Sigma}_n$ to Σ, and continuity of the characteristic polynomial then forces the sample eigenvalues to converge to the population eigenvalues. The size and shape of the fluctuations around these limits depend on whether the population eigenvalues are simple or repeated. For $\Sigma = I_p$ the matrix $\sqrt{n}(\widehat{\Sigma}_n - I_p)$ converges in distribution to a *Gaussian Orthogonal Ensemble* matrix, whose top eigenvalue has a non-Gaussian limit shaped by *eigenvalue repulsion*. For Σ with distinct eigenvalues, the delta method instead produces ordinary Gaussian fluctuations of each sample eigenvalue around its population counterpart.

2. *The high-dimensional regime* ($p, n \to \infty$ with $p/n \to y \in (0, \infty)$). Even for the null covariance $\Sigma = I_p$, the sample eigenvalues no longer concentrate at 1; they fill out the *Marchenko-Pastur law* on the interval $[(1-\sqrt{y})^2, (1+\sqrt{y})^2]$ (Marchenko and Pastur,

1967), exactly as in the right panel of Figure 1. The largest eigenvalue converges almost surely to the upper edge $(1+\sqrt{y})^2$, but its fluctuations around that edge are not Gaussian. They live on the scale $p^{-2/3}$ and follow the *Tracy-Widom law* (Tracy and Widom, 1996; Johnstone, 2001), a distinct universality class first discovered in random matrix theory and now found in problems ranging from longest increasing subsequences to growth models in mathematical physics.

3. *Structure and a phase transition.* When the population covariance carries low-rank structure, a small number of distinguished eigenvalues riding above a flat baseline, a striking new phenomenon appears. If a signal eigenvalue exceeds a threshold determined by the aspect ratio y, an isolated outlier eigenvalue appears in the sample spectrum and the corresponding sample eigenvector aligns nontrivially with the population spike direction. If the signal is below the threshold, the spike is absorbed into the Marchenko-Pastur bulk and its eigenvector becomes asymptotically orthogonal to the population direction. This phase transition is the bridge to spectral clustering: the same picture explains how clustering algorithms recover hidden class structure from data, but only when the underlying signal-to-noise ratio crosses an analogous threshold.

Where this is applicable

The techniques developed here are not abstract curiosities. The Marchenko-Pastur law sets a baseline for detecting structure in any high-dimensional covariance estimate, with applications in array-signal processing, wireless communications, and quantitative finance (Couillet and Liao, 2022). The Tracy-Widom edge governs the null distribution of the largest principal component, which is the basis for principled testing of how many genuine signal directions are present in a data set (Johnstone, 2001). The spike phase transition determines when a low-rank signal in noise is recoverable by PCA, with direct implications for denoising and dimensionality reduction. Spectral clustering, finally, is one of the most widely used unsupervised methods for problems where the natural geometry is not Euclidean, including image segmentation, community detection in networks, and clustering of single-cell data (von Luxburg, 2007; Ng et al., 2002). The phase transition we develop in the spiked model is the cleanest mathematical lens we have for understanding when these methods succeed and when they fail.

Roadmap

Section 2 reviews probability foundations: random variables, moments and moment generating functions, characteristic functions, modes of convergence, the law of large numbers and central limit theorem, and random vectors. Section 3 develops the sample covariance matrix in both regimes, from the spectral theorem and the Wishart distribution in low dimensions through the Marchenko-Pastur and Tracy-Widom laws in high dimensions. Section 4 introduces the spiked covariance model, proves the eigenvalue phase transition, derives the eigenvector overlap formula, and sketches the connection to spectral clustering. Section 5 concludes with a summary and open questions, and three appendices collect auxiliary probabilistic results, simulation methodology, and selected Python code listings.

2 Probability Foundations

The high-dimensional phenomena studied in later sections all rest on a small set of foundational ideas from classical probability: distributions of random variables, ways to summarize and compare them, modes of convergence, the law of large numbers, and the central limit theorem. This section develops these tools in the depth needed to make later proofs go through cleanly. The exposition starts with the random variable, the basic object of probability theory, and builds toward the multivariate normal distribution that will drive every Gaussian sample-covariance computation in Section 3.

2.1 Random Variables and Distributions

Probability begins with a *probability space* $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the set of possible outcomes, $\mathcal{F}$ is a collection of events, and $\mathbb{P}$ assigns each event a number in $[0, 1]$. The full development imposes some technical regularity on $\mathcal{F}$ and $\mathbb{P}$ (the language of σ-algebras and measures, developed in measure-theoretic probability); we will not need to engage with it directly. We work with this triple in the background; the foreground objects are real-valued functions of the outcome.

Definition 2.1 (Random variable)**.** A *random variable* is a function $X : \Omega \to \mathbb{R}$. Strictly, X is required to satisfy the technical condition $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$, so that $\mathbb{P}(X \leq x)$ is well defined; this is called *measurability*, holds for every random variable we will encounter, and we will not verify it explicitly. The distribution of X is captured by the *cumulative distribution function* (CDF)

$$F_X(x) \;=\; \mathbb{P}(X \leq x), \qquad x \in \mathbb{R}.$$

A CDF is determined by three properties.

Proposition 2.1 (Characterization of CDFs)**.** The CDF of any random variable satisfies:

(F1) $\lim\limits_{x \to -\infty} F_X(x) = 0$ and $\lim\limits_{x \to +\infty} F_X(x) = 1$;

(F2) F_X is right-continuous: $\lim_{x \to a^+} F_X(x) = F_X(a)$;

(F3) F_X is non-decreasing: $a \leq b \Rightarrow F_X(a) \leq F_X(b)$.

Conversely, any $F : \mathbb{R} \to [0, 1]$ satisfying (F1)–(F3) is the CDF of some random variable (Casella and Berger, 2002, §1.5).

Random variables come in two main flavors. *Discrete* random variables take values in a countable set $\{x_1, x_2, \ldots\}$, and their distribution is described by the *probability mass function* (PMF) $p_X(x_i) = \mathbb{P}(X = x_i)$. *Absolutely continuous* random variables admit a nonnegative *probability density function* (PDF) f_X with

$$F_X(x) \;=\; \int_{-\infty}^{x} f_X(t)\, dt.$$

Two random variables X, Y are *independent* if $\mathbb{P}(X \in A,\, Y \in B) = \mathbb{P}(X \in A)\,\mathbb{P}(Y \in B)$ for all intervals $A, B \subseteq \mathbb{R}$. (More generally one asks the identity to hold on all events; the version on intervals suffices for real-valued random variables.)

Example 2.2 (Distributions used throughout)**.** The following appear repeatedly in later sections:

- *Bernoulli*(p): $X \in \{0, 1\}$ with $\mathbb{P}(X = 1) = p$.
- *Poisson*(λ): $\mathbb{P}(X = k) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$.
- *Normal* $\mathcal{N}(\mu, \sigma^2)$: $f_X(x) = (\sqrt{2\pi}\,\sigma)^{-1} \exp\big(-(x-\mu)^2/(2\sigma^2)\big)$.
- *Exponential*(λ): $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.
- *Cauchy*: $f_X(x) = 1/\big(\pi(1 + x^2)\big)$.

Their shapes are shown in Figure 2.

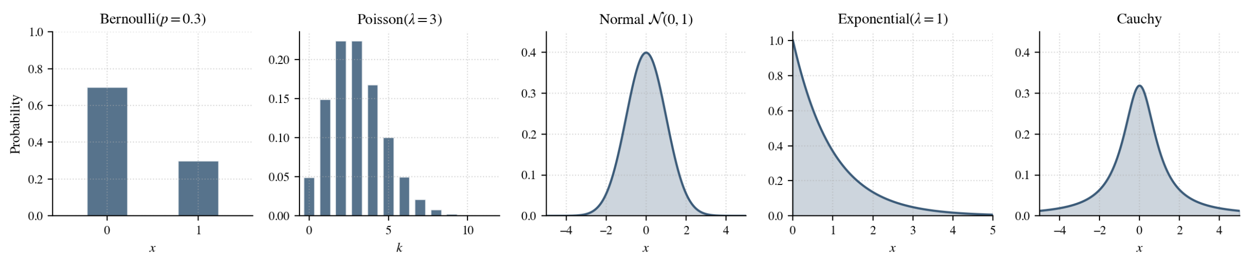

Figure 2: Shapes of the five distributions used throughout the report. Bernoulli and Poisson are discrete, displayed as probability mass functions; Normal, Exponential, and Cauchy are absolutely continuous, displayed as densities. The Normal and the Cauchy share a similar central profile but very different tails: the Cauchy density decays only like $1/x^2$, so values far from zero remain plausible. This tail behavior is what breaks the moment generating function in Section 2.2 and motivates the characteristic function in Section 2.3.

The Cauchy distribution will be a recurring foil: its heavy tails make some standard tools break, motivating sharper tools that do not.

2.2 Moments and Moment Generating Functions

Moments quantify the shape of a distribution.

Definition 2.3 (Moments and central moments)**.** For an integer $j \geq 1$ such that $\mathbb{E}|X|^j < \infty$, the j^{th} *moment* of X is $\mathbb{E}[X^j]$ and the j^{th} *central moment* is $\mathbb{E}[(X - \mu)^j]$, where $\mu = \mathbb{E}[X]$. The second central moment is the *variance*, $\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2]$. The absolute-moment hypothesis ensures that X^j is integrable, so the bare expectation $\mathbb{E}[X^j]$ is well defined.

A generating function for the moments compresses all of this information into a single function.

Definition 2.4 (Moment generating function)**.** The *moment generating function* (MGF) of X is

$$M_X(t) \;=\; \mathbb{E}[e^{tX}], \qquad t \in \mathbb{R},$$

provided the expectation is finite on an open neighborhood $(-h, h)$ of zero.

The name is justified by the following.

Theorem 2.2 (MGFs generate moments)**.** If $M_X(t)$ is finite for $t \in (-h, h)$, then M_X is infinitely differentiable at 0 and

$$M_X^{(n)}(0) \;=\; \mathbb{E}[X^n], \qquad n = 1, 2, \ldots$$

Proof. We treat the continuous case in detail; the discrete case is identical with sums in place of integrals. The hypothesis that M_X is finite on an open interval around zero is the standard regularity condition that justifies differentiating under the integral sign in the calculation that follows; see Casella and Berger (2002, Thm. 2.3.7). Therefore,

$$\frac{d}{dt} M_X(t) \;=\; \frac{d}{dt} \int_{-\infty}^{\infty} e^{tx} f_X(x)\, dx \;=\; \int_{-\infty}^{\infty} \frac{d}{dt}\big(e^{tx}\big) f_X(x)\, dx.$$

The inner derivative is $\frac{d}{dt} e^{tx} = x\, e^{tx}$, so

$$\frac{d}{dt} M_X(t) \;=\; \int_{-\infty}^{\infty} x\, e^{tx} f_X(x)\, dx \;=\; \mathbb{E}\big[X\, e^{tX}\big].$$

Setting $t = 0$ inside the expectation gives $e^{0 \cdot X} = 1$, and hence

$$M_X'(0) \;=\; \mathbb{E}\big[X\, e^{0 \cdot X}\big] \;=\; \mathbb{E}[X].$$

Iterating the same step n times produces

$$M_X^{(n)}(t) \;=\; \mathbb{E}\big[X^n\, e^{tX}\big],$$

and evaluating at $t = 0$ yields $M_X^{(n)}(0) = \mathbb{E}[X^n]$, as claimed.

In the discrete case, the same calculation applies to the series $M_X(t) = \sum_x e^{tx} p_X(x)$; convergence of M_X on $(-h, h)$ justifies term-by-term differentiation, and the rest of the argument is unchanged. $\square$

A second property is the source of most computational power: the MGF of a sum of independent random variables is the product of their MGFs. Indeed, if X and Y are independent,

$$M_{X+Y}(t) \;=\; \mathbb{E}[e^{t(X+Y)}] \;=\; \mathbb{E}[e^{tX}]\, \mathbb{E}[e^{tY}] \;=\; M_X(t)\, M_Y(t).$$

We compute MGFs of the canonical distributions to make this concrete.

Example 2.5 (MGFs of standard distributions)**.** We compute MGFs from the definition for three of the distributions in Example 2.2.

Bernoulli(p). The variable X takes only the values 0 and 1, so the expectation is a two-term sum:

$$M_X(t) \;=\; \mathbb{E}[e^{tX}] \;=\; e^{t\cdot 0}\,\mathbb{P}(X=0) \;+\; e^{t\cdot 1}\,\mathbb{P}(X=1) \;=\; (1-p) \;+\; p\,e^t.$$

Poisson(λ). Substitute the PMF $\mathbb{P}(X=k) = e^{-\lambda}\lambda^k/k!$ into the expectation:

$$M_X(t) \;=\; \sum_{k=0}^{\infty} e^{tk}\,e^{-\lambda}\,\frac{\lambda^k}{k!} \;=\; e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda\,e^t)^k}{k!}.$$

The inner sum is the Taylor expansion of exp evaluated at λe^t, namely $\exp(\lambda e^t)$. Combining,

$$M_X(t) \;=\; e^{-\lambda}\,e^{\lambda e^t} \;=\; \exp\bigl(\lambda(e^t-1)\bigr).$$

Normal $\mathcal{N}(\mu,\sigma^2)$. Starting from the density,

$$M_X(t) \;=\; \frac{1}{\sqrt{2\pi}\,\sigma}\int_{-\infty}^{\infty}\exp\Bigl(tx \;-\; \frac{(x-\mu)^2}{2\sigma^2}\Bigr)\,dx.$$

The trick is to complete the square in the exponent so the integrand becomes a Gaussian kernel. Expanding the quadratic and collecting terms,

$$tx \;-\; \frac{(x-\mu)^2}{2\sigma^2} \;=\; -\frac{1}{2\sigma^2}\Bigl(x-(\mu+\sigma^2 t)\Bigr)^2 \;+\; \Bigl(\mu t \;+\; \tfrac{1}{2}\sigma^2 t^2\Bigr).$$

The first piece is the kernel of a normal density with mean $\mu+\sigma^2 t$ and variance σ^2; divided by $\sqrt{2\pi}\,\sigma$ it integrates to 1. The second piece does not depend on x and comes outside the integral. Therefore,

$$M_X(t) \;=\; \exp\Bigl(\mu t \;+\; \tfrac{1}{2}\sigma^2 t^2\Bigr), \qquad t\in\mathbb{R}.$$

These three examples illustrate the technique. The multiplicative property turns difficult convolutions into easy multiplications: if $X_1,\dots,X_n$ are iid Bernoulli(p), their sum has MGF $(1-p+pe^t)^n$, which is the MGF of Binomial(n,p); independent Poissons of rates λ_1,λ_2 sum to a Poisson of rate $\lambda_1+\lambda_2$; independent normals sum to a normal with summed mean and summed variance.

Beyond computing moments, the MGF determines the distribution uniquely.

Theorem 2.3 (MGF uniqueness)**.** Let X and Y be real-valued random variables.

(1) If X and Y have the same distribution and $M_X(t)$ is finite at t, then $M_X(t) = M_Y(t)$.

(2) Conversely, if $M_X(t) = M_Y(t)$ for all t in some open neighborhood $(-h,h)$ of zero, then X and Y have the same distribution.

Proof of (1). If X and Y have the same distribution, then for any function g for which both expectations are well defined,

$$\mathbb{E}[g(X)] \;=\; \mathbb{E}[g(Y)].$$

(Both sides are computed by the same integral or sum against the common distribution function, so they agree.) Choosing $g(x) = e^{tx}$, which by hypothesis has a finite expectation under X at t, yields

$$M_X(t) \;=\; \mathbb{E}[e^{tX}] \;=\; \mathbb{E}[e^{tY}] \;=\; M_Y(t),$$

which is the claim. □

The converse direction (2) is the harder one: it says that the entire distribution of X is encoded in the MGF on any open interval around zero. The proof reduces to the uniqueness of the two-sided Laplace transform, since $M_X(t)$ is exactly that transform of the distribution evaluated at $s = -t$; see Casella and Berger (2002, Thm. 2.3.11) for the details.

There is a catch. The MGF requires e^{tX} to have a finite expectation for t near zero, which can fail if X has heavy tails.

Example 2.6 (Cauchy: no MGF)**.** For the standard Cauchy density $f(x) = 1/(\pi(1 + x^2))$, the MGF integral

$$\int_{-\infty}^{\infty} \frac{e^{tx}}{\pi(1+x^2)}\,dx$$

diverges for every $t \neq 0$, since the integrand grows like e^{tx}/x^2 at the right tail (for $t > 0$) or at the left tail (for $t < 0$). The MGF therefore does not exist on any open neighborhood of zero. This is not a pathology; the Cauchy is a perfectly well-defined distribution, central in probability and physics.

The remedy is the characteristic function.

2.3 Characteristic Functions

Replacing the real argument t of the MGF by the imaginary argument it yields an object that always exists.

Definition 2.7 (Characteristic function)**.** The *characteristic function* (CF) of a random variable X is

$$\varphi_X(t) \;=\; \mathbb{E}[e^{itX}] \;=\; \mathbb{E}[\cos(tX)] + i\,\mathbb{E}[\sin(tX)], \qquad t \in \mathbb{R}.$$

Because $|e^{itx}| = 1$ for every real x, the integrand is bounded, and so

$$|\varphi_X(t)| \;\le\; \mathbb{E}[|e^{itX}|] \;=\; 1, \qquad \varphi_X(0) = 1.$$

The CF is therefore well defined for every distribution. When X has a density, φ_X is exactly the Fourier transform of the density:

$$\varphi_X(t) \;=\; \int_{-\infty}^{\infty} e^{itx}\, f_X(x)\,dx.$$

When the MGF exists, formal substitution gives $\varphi_X(t) = M_X(it)$.

Example 2.8 (Standard normal CF)**.** For $X \sim \mathcal{N}(0,1)$, the MGF computed in Example 2.5 is

$$M_X(s) \;=\; e^{s^2/2},$$

and this expression is well defined for every complex s. Substituting $s = it$ using the identity $\varphi_X(t) = M_X(it)$ yields

$$\varphi_X(t) \;=\; M_X(it) \;=\; e^{(it)^2/2} \;=\; e^{-t^2/2}, \qquad t \in \mathbb{R}. \tag{2}$$

A direct computation from the definition of φ_X, completing the square in the exponent of the integrand, gives the same answer.

Example 2.9 (Standard Cauchy CF)**.** Let X be standard Cauchy with density $f(x) = 1/(\pi(1+x^2))$. By definition,

$$\varphi_X(t) \;=\; \int_{-\infty}^{\infty} e^{itx}\,\frac{1}{\pi(1+x^2)}\,dx \;=\; \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{e^{itx}}{1+x^2}\,dx.$$

The remaining integral is a standard Fourier-transform identity (Feller (1971, §XV.4); or any Fourier-transform table):

$$\int_{-\infty}^{\infty}\frac{e^{itx}}{1+x^2}\,dx \;=\; \pi\, e^{-|t|}, \qquad t \in \mathbb{R}.$$

Dividing by π,

$$\varphi_X(t) \;=\; e^{-|t|}, \qquad t \in \mathbb{R}. \tag{3}$$

The Cauchy CF is well defined and continuous, even though the Cauchy MGF is identically infinite for every $t \neq 0$.

Figure 3 contrasts the two cases. Although the Cauchy and Normal densities look superficially similar near the origin, their tails are very different, and that difference is exactly what the characteristic function transports into a well-behaved object.

Two algebraic properties make CFs computationally convenient. By independence, $\varphi_{X+Y}(t) = \varphi_X(t)\,\varphi_Y(t)$; by direct substitution, $\varphi_{aX+b}(t) = e^{itb}\,\varphi_X(at)$. Like the MGF, the CF determines the distribution uniquely, and an inversion formula recovers the CDF from φ_X (Billingsley, 1995, §26).

The deepest property of characteristic functions is the bridge they provide between pointwise convergence of φ_{X_n} and convergence in distribution of X_n.

Theorem 2.4 (Lévy continuity theorem)**.** Let X_n and X be random variables with characteristic functions φ_{X_n} and φ_X.

(1) If $X_n \xrightarrow{d} X$, then $\varphi_{X_n}(t) \to \varphi_X(t)$ for every $t \in \mathbb{R}$.

(2) Conversely, if $\varphi_{X_n}(t) \to \varphi(t)$ for every $t \in \mathbb{R}$ and φ is continuous at 0, then φ is the CF of some random variable X, and $X_n \xrightarrow{d} X$.

A full proof is in Billingsley (1995, Thm. 26.3). The theorem is the key to the central limit theorem proof in Section 2.5: it reduces a question about distributions to a calculus exercise on a single complex-valued function.

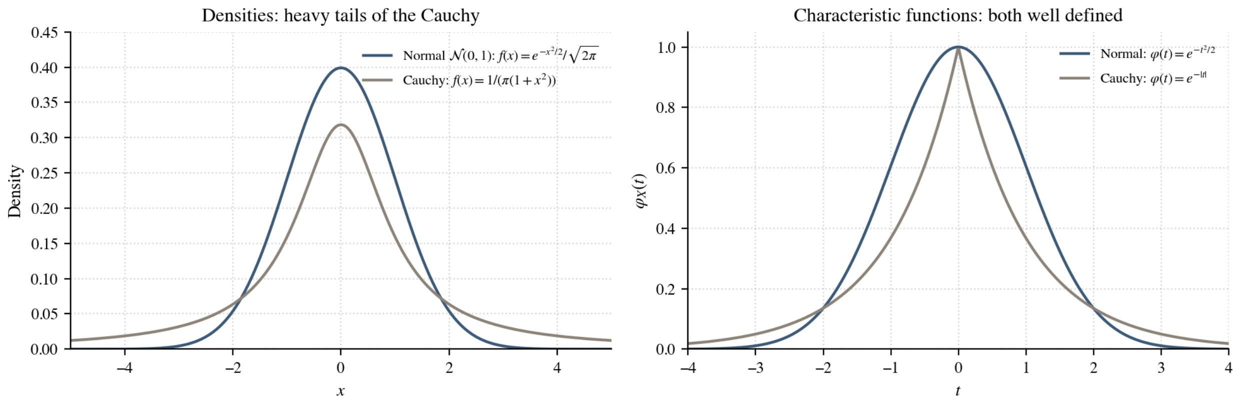

Figure 3: Left: the standard Normal and standard Cauchy densities on the same axes. The Cauchy has a slightly lower peak and noticeably heavier tails; the area in the tails beyond $|x| = 3$ is several times larger for the Cauchy than for the Normal. Right: their characteristic functions, $e^{-t^2/2}$ for the Normal and $e^{-|t|}$ for the Cauchy. Both are well-defined real functions on all of $\mathbb{R}$ and decay to zero, even though the Cauchy's MGF is identically infinite for every $t \neq 0$. This is the practical sense in which the CF rescues the heavy-tailed case.

2.4 Modes of Convergence

A sequence of random variables can converge to a limit in several inequivalent ways. The three modes used in this report, from strongest to weakest, are the following.

Definition 2.10 (Three modes of convergence)**.** Let X_n, X be random variables on a common probability space.

- $X_n \xrightarrow{\text{a.s.}} X$ (*almost surely*): $\mathbb{P}\big(\lim_n X_n = X\big) = 1$.
- $X_n \xrightarrow{\mathbb{P}} X$ (*in probability*): for every $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$.
- $X_n \xrightarrow{d} X$ (*in distribution*): $F_{X_n}(x) \to F_X(x)$ at every continuity point of F_X.

Proposition 2.5 (Implication chain)**.** $X_n \xrightarrow{\text{a.s.}} X \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X$. The converses are false in general.

Sketch. For a.s. $\Rightarrow$ in probability: pointwise convergence holds with probability one, so for every $\varepsilon > 0$ the probability that $|X_n - X| > \varepsilon$ tends to zero. For in probability $\Rightarrow$ in distribution: the conclusion is on CDFs at continuity points and follows by a standard ε-argument; see Casella and Berger (2002, Thm. 5.5.12). □

A canonical counterexample showing that probability does not imply almost-sure convergence:

Example 2.11 ($\xrightarrow{\mathbb{P}}$ but not $\xrightarrow{\text{a.s.}}$)**.** Define a sequence $X_n \in \{0, 1\}$ by

$$\mathbb{P}(X_n = 1) \;=\; \frac{1}{n}, \qquad \mathbb{P}(X_n = 0) \;=\; 1 - \frac{1}{n},$$

and assume the X_n are independent.

Convergence in probability to 0. Fix any $\varepsilon \in (0,1)$. Since X_n only takes the values 0 and 1, the event $\{|X_n - 0| > \varepsilon\}$ is exactly $\{X_n = 1\}$. Therefore,

$$\mathbb{P}(|X_n - 0| > \varepsilon) \;=\; \mathbb{P}(X_n = 1) \;=\; \frac{1}{n} \;\longrightarrow\; 0,$$

which is the definition of $X_n \xrightarrow{\mathbb{P}} 0$.

Failure of almost-sure convergence. Let $A_n = \{X_n = 1\}$. The events A_n are independent and their probabilities satisfy

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) \;=\; \sum_{n=1}^{\infty} \frac{1}{n} \;=\; \infty.$$

The second Borel-Cantelli lemma (Appendix A) says that for independent events whose probabilities sum to infinity, infinitely many occur with probability one. Concretely,

$$\mathbb{P}\big(X_n = 1 \text{ for infinitely many } n\big) \;=\; 1.$$

So almost every realization of the sequence keeps producing the value 1 at arbitrarily late indices, which prevents it from settling at 0. Hence $X_n \overset{\text{a.s.}}{\not\longrightarrow} 0$.

A second standard fact will be useful in later sections: continuous functions preserve convergence.

Proposition 2.6 (Continuous mapping theorem)**.** If $X_n \xrightarrow{\text{a.s.}} X$ (respectively $\xrightarrow{\mathbb{P}}$, $\xrightarrow{d}$) and $g : \mathbb{R} \to \mathbb{R}$ is continuous, then $g(X_n) \xrightarrow{\text{a.s.}} g(X)$ (respectively $\xrightarrow{\mathbb{P}}$, $\xrightarrow{d}$).

Slutsky's theorem (also useful later) is recorded in Appendix A.

2.5 Limit Theorems: LLN and CLT

Two universality results anchor classical probability: averages of iid samples converge to their mean (law of large numbers), and properly normalized averages have a universal Gaussian fluctuation (central limit theorem). We prove both, building on the tools of Sections 2.3–2.4.

We first need two classical inequalities.

Theorem 2.7 (Markov's inequality)**.** If $Y \geq 0$ and $a > 0$, then $\mathbb{P}(Y \geq a) \leq \mathbb{E}[Y]/a$.

Proof. On the event $\{Y \geq a\}$, the variable Y itself satisfies $Y \geq a$. Multiplying both sides by the indicator $\mathbf{1}_{\{Y \geq a\}}$,

$$Y\,\mathbf{1}_{\{Y \geq a\}} \;\geq\; a\,\mathbf{1}_{\{Y \geq a\}}.$$

Outside the event both sides equal zero, so the inequality holds everywhere on Ω. Take expectations and use that $\mathbf{1}_{\{Y \geq a\}}$ integrates to $\mathbb{P}(Y \geq a)$:

$$\mathbb{E}\big[Y\,\mathbf{1}_{\{Y \geq a\}}\big] \;\geq\; a\,\mathbb{P}(Y \geq a).$$

Finally, $Y\,\mathbf{1}_{\{Y\geq a\}} \leq Y$ (because $Y \geq 0$ and the indicator is at most 1), so

$$\mathbb{E}[Y] \;\geq\; \mathbb{E}\big[Y\,\mathbf{1}_{\{Y\geq a\}}\big] \;\geq\; a\,\mathbb{P}(Y \geq a).$$

Dividing by $a > 0$ gives the claim. □

Theorem 2.8 (Chebyshev's inequality)**.** If $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2 < \infty$, then for any $\varepsilon > 0$,

$$\mathbb{P}(|X-\mu| \geq \varepsilon) \;\leq\; \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Squaring is strictly increasing on $[0,\infty)$, so the event $\{|X-\mu| \geq \varepsilon\}$ coincides with $\{(X-\mu)^2 \geq \varepsilon^2\}$:

$$\mathbb{P}(|X-\mu| \geq \varepsilon) \;=\; \mathbb{P}\big((X-\mu)^2 \geq \varepsilon^2\big).$$

The variable $Y = (X-\mu)^2$ is nonnegative with $\mathbb{E}[Y] = \mathrm{Var}(X) = \sigma^2$. Applying Markov's inequality (Theorem 2.7) to Y with threshold $a = \varepsilon^2$,

$$\mathbb{P}\big((X-\mu)^2 \geq \varepsilon^2\big) \;\leq\; \frac{\mathbb{E}[(X-\mu)^2]}{\varepsilon^2} \;=\; \frac{\sigma^2}{\varepsilon^2}.$$

Combining the two displays gives the claim. □

These suffice to prove the weak law.

Theorem 2.9 (Chebyshev's weak law of large numbers)**.** Let $X_1, X_2, \ldots$ be iid with $\mathbb{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 < \infty$. Let $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$. Then $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. Compute the mean and variance of $\bar{X}_n$. By linearity of expectation,

$$\mathbb{E}[\bar{X}_n] \;=\; \frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i] \;=\; \mu.$$

By independence of the X_i,

$$\mathrm{Var}(\bar{X}_n) \;=\; \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(X_i) \;=\; \frac{\sigma^2}{n}.$$

Apply Chebyshev's inequality (Theorem 2.8) to $\bar{X}_n$ with threshold ε:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \;\leq\; \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} \;=\; \frac{\sigma^2}{n\varepsilon^2} \;\longrightarrow\; 0 \qquad (n\to\infty).$$

This is the definition of $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$. □

A stronger statement holds without the finite-variance assumption.

Theorem 2.10 (Kolmogorov's strong law of large numbers)**.** If $X_1, X_2, \ldots$ are iid with $\mathbb{E}[|X_1|] < \infty$ and $\mathbb{E}[X_1] = \mu$, then $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$.

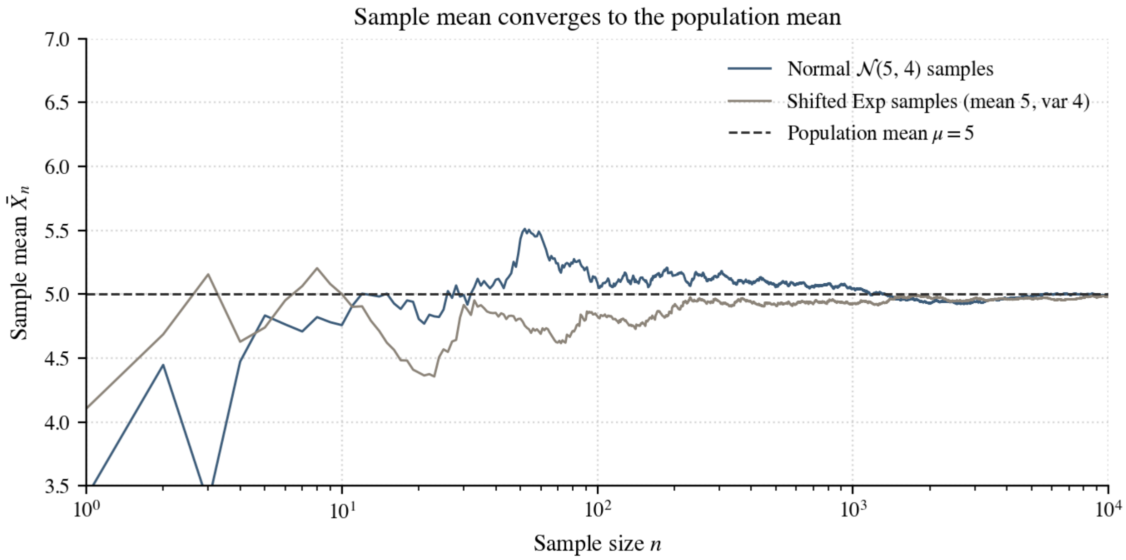

Figure 4: The sample mean $\bar{X}_n$ converges to the population mean $\mu = 5$ for two distributions with very different shapes but the same mean and variance: a normal $\mathcal{N}(5, 4)$ and an exponential rescaled and shifted to also have mean 5 and variance 4. The exponential samples fluctuate more in the early stages because their distribution is heavily skewed, but both averages settle near 5 by $n = 10000$. The choice of normal vs. shifted-exponential comparison with matched mean and variance follows Figure 1 of Genzer (2025); the figure here is generated by `notebooks/01_lln_clt_demos.py`.

A proof is given in Billingsley (1995, Thm. 22.1). Figure 4 shows the convergence concretely.

The central limit theorem refines the law of large numbers by saying *at what rate* the convergence happens, and *what the fluctuations look like.*

Theorem 2.11 (Central limit theorem)**.** Let $X_1, X_2, \dots$ be iid with $\mathbb{E}[X_1] = \mu$ and $0 < \mathrm{Var}(X_1) = \sigma^2 < \infty$. Then

$$Z_n \;:=\; \frac{\sqrt{n}\,(\bar{X}_n - \mu)}{\sigma} \;\xrightarrow{d}\; \mathcal{N}(0, 1).$$

Proof. The proof has four steps: standardize, expand the characteristic function near zero, factor across the iid sum, and pass to the limit through Lévy's continuity theorem.

Step 1 (standardize). Set $Y_i = (X_i - \mu)/\sigma$. Then $\mathbb{E}[Y_i] = 0$, $\mathrm{Var}(Y_i) = 1$, and

$$Z_n \;=\; \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i.$$

Step 2 (expand the CF near zero). Let $\varphi(t) = \mathbb{E}[e^{itY_1}]$. The Taylor expansion $e^{iu} = 1 + iu - u^2/2 + o(u^2)$ as $u \to 0$, applied with $u = tY_1$, gives

$$e^{itY_1} \;=\; 1 \;+\; itY_1 \;-\; \tfrac{1}{2}t^2Y_1^2 \;+\; o(t^2Y_1^2).$$

Taking expectations and using $\mathbb{E}[Y_1] = 0$ and $\mathbb{E}[Y_1^2] = 1$,

$$\varphi(t) \;=\; 1 \;-\; \tfrac{1}{2}t^2 \;+\; o(t^2) \qquad (t \to 0).$$

Step 3 (factor by independence). The Y_i are independent, so the characteristic function of Z_n factors into a product of n copies of $\varphi(t/\sqrt{n})$:

$$\varphi_{Z_n}(t) \;=\; \mathbb{E}\!\left[\exp\!\Big(\frac{it}{\sqrt{n}} \sum_{i=1}^{n} Y_i\Big)\right] \;=\; \prod_{i=1}^{n} \mathbb{E}\big[e^{i(t/\sqrt{n})Y_i}\big] \;=\; \Big(\varphi\big(t/\sqrt{n}\big)\Big)^{n}.$$

Substituting the expansion from Step 2 with $t/\sqrt{n}$ in place of t,

$$\varphi_{Z_n}(t) \;=\; \Big(1 \;-\; \tfrac{t^2}{2n} \;+\; o(1/n)\Big)^{n}.$$

Step 4 (take the limit and conclude). The standard calculus identity $(1 + a_n/n)^n \to e^a$ when $a_n \to a$, applied with $a_n \to -t^2/2$, gives

$$\varphi_{Z_n}(t) \;\longrightarrow\; e^{-t^2/2} \qquad (n \to \infty).$$

By Example 2.8, $e^{-t^2/2}$ is the characteristic function of $\mathcal{N}(0,1)$. The Lévy continuity theorem (Theorem 2.4) then yields $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. $\square$

Figure 5 illustrates the CLT for exponential samples: even though the underlying distribution is sharply skewed, the histogram of the standardized sample mean Z_n approaches the standard normal density as n grows.

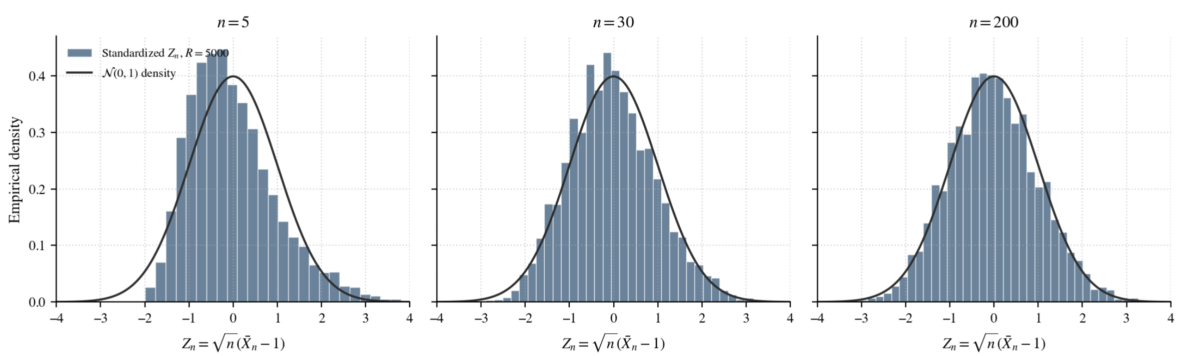

Figure 5: Central limit theorem for iid Exponential(1) samples. Each panel shows a histogram of $R = 5000$ standardized sample means $Z_n = \sqrt{n}(\bar{X}_n - 1)$ at increasing n. The histogram is sharply skewed at $n = 5$, visibly less so at $n = 30$, and indistinguishable from the standard normal density (solid line) by $n = 200$. Generated by `notebooks/01_lln_clt_demos.py`.

2.6 Random Vectors and the Multivariate Normal Distribution

The objects of this report are sample covariance matrices, which act on random vectors rather than random scalars. This subsection collects the multivariate analogues of Sections 2.1–2.5: random vectors and joint distributions, the covariance matrix and its first structural property, the multivariate normal distribution, and the multivariate central limit theorem. The linear-algebra side of the story (spectral theorem, eigenvalue characterization of positive semidefiniteness, norms, and the convergence of eigenvalues under matrix convergence) is collected in Section 2.7.

Random vectors and joint distributions.

Definition 2.12 (Random vector and joint CDF)**.** A *random vector* is a function $\boldsymbol{X} = (X_1, \dots, X_p)^\top : \Omega \to \mathbb{R}^p$ whose components are random variables on a common probability space. The *joint CDF* of $\boldsymbol{X}$ is

$$F_{\boldsymbol{X}}(\boldsymbol{x}) \;=\; \mathbb{P}(X_1 \le x_1, \, \dots, \, X_p \le x_p), \qquad \boldsymbol{x} \in \mathbb{R}^p.$$

The joint CDF obeys the multivariate analogues of properties (F1)–(F3) of Proposition 2.1.

Proposition 2.12 (Properties of the joint CDF)**.** The joint CDF $F_{\boldsymbol{X}} : \mathbb{R}^p \to [0,1]$ satisfies:

(F1) As $x_i \to -\infty$ for any single coordinate i, $F_{\boldsymbol{X}}(\boldsymbol{x}) \to 0$; as all coordinates tend to $+\infty$, $F_{\boldsymbol{X}}(\boldsymbol{x}) \to 1$.

(F2) $F_{\boldsymbol{X}}$ is right-continuous in each coordinate.

(F3) $F_{\boldsymbol{X}}$ is non-decreasing in each coordinate.

(F4) Marginal CDFs are recovered by limits: $F_{X_1}(x_1) = \lim_{x_2,\dots,x_p \to +\infty} F_{\boldsymbol{X}}(x_1, \dots, x_p)$, and similarly for other coordinates.

Moreover, $X_1, \dots, X_p$ are independent if and only if $F_{\boldsymbol{X}}(\boldsymbol{x}) = \prod_{i=1}^p F_{X_i}(x_i)$ for every $\boldsymbol{x} \in \mathbb{R}^p$.

The discrete and absolutely continuous cases admit joint PMFs and joint densities, defined exactly as in the scalar case.

Three examples of joint distributions. The next three examples illustrate the spectrum from full independence to maximal dependence to a smooth Gaussian dependence. They also show the difference between the *support* of a random vector (where it lives) and its *distribution* (how its mass is spread on its support).

Example 2.13 (Independent exponentials)**.** Let X_1, X_2 be iid with $X_i \sim \text{Exponential}(\lambda)$. Each scalar CDF is $F_X(x) = 1 - e^{-\lambda x}$ for $x \ge 0$ (and 0 for $x < 0$). By independence,

$$F_{X_1,X_2}(x_1, x_2) \;=\; F_X(x_1)\, F_X(x_2) \;=\; (1 - e^{-\lambda x_1})(1 - e^{-\lambda x_2}), \qquad x_1, x_2 \ge 0,$$

and $F_{X_1,X_2}(x_1, x_2) = 0$ if either coordinate is negative. The mass lives on the positive quadrant $\{x_1 \ge 0, \, x_2 \ge 0\}$, and the joint density factors as $f_{X_1,X_2} = f_{X_1} f_{X_2}$.

Example 2.14 (Maximally dependent: $X_1 = X_2$ a.s.)**.** Let $X_1 \sim \text{Exponential}(\lambda)$ and set $X_2 = X_1$ almost surely. Then $\mathbb{P}(X_1 = X_2) = 1$, so the random point (X_1, X_2) lies on the diagonal $\{x_1 = x_2 \geq 0\}$ with probability one. The joint CDF is

$$\begin{aligned} F_{X_1,X_2}(x_1, x_2) \;&=\; \mathbb{P}(X_1 \leq x_1,\, X_2 \leq x_2) \\ &=\; \mathbb{P}(X_1 \leq \min\{x_1, x_2\}) \;=\; F_X(\min\{x_1, x_2\}). \end{aligned}$$

The two components have the same one-dimensional marginals as in Example 2.13, but the support of the pair collapses to a one-dimensional diagonal ray. This vector does *not* admit a joint density on $\mathbb{R}^2$, because all its mass is concentrated on a set of Lebesgue area zero. We will see shortly that it is the prototypical case of a covariance matrix that is positive semidefinite but *not* positive definite.

Example 2.15 (Bivariate normal density)**.** Let $\boldsymbol{X} = (X_1, X_2)^\top \sim \mathcal{N}(\boldsymbol{0}, \Sigma)$ with

$$\Sigma \;=\; \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \qquad \sigma_1, \sigma_2 > 0, \quad -1 < \rho < 1.$$

A short computation gives $\det \Sigma = \sigma_1^2\sigma_2^2(1 - \rho^2)$ and

$$\Sigma^{-1} \;=\; \frac{1}{\sigma_1^2\sigma_2^2(1 - \rho^2)} \begin{pmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{pmatrix}.$$

Substituting into the multivariate normal density (Definition 2.17 below) and simplifying yields the familiar scalar formula

$$f_{X_1,X_2}(x_1, x_2) \;=\; \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1 - \rho^2}} \exp\!\left(-\frac{1}{2(1 - \rho^2)}\left[\frac{x_1^2}{\sigma_1^2} \;-\; \frac{2\rho x_1 x_2}{\sigma_1\sigma_2} \;+\; \frac{x_2^2}{\sigma_2^2}\right]\right).$$

When $\rho = 0$ the cross term vanishes and the density factors as $f_{X_1} f_{X_2}$, so $X_1 \perp X_2$. When $\rho \neq 0$ the cross term cannot be removed and the components are dependent.

Figure 6 contrasts the three cases at a glance.

Mean vector and covariance matrix.

Definition 2.16 (Mean vector and covariance matrix)**.** The *mean vector* of $\boldsymbol{X}$ is $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{X}] = (\mathbb{E}[X_1], \ldots, \mathbb{E}[X_p])^\top$. The *covariance matrix* is

$$\Sigma \;=\; \operatorname{Cov}(\boldsymbol{X}) \;=\; \mathbb{E}\big[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\big],$$

a $p \times p$ matrix with entries

$$\Sigma_{ij} \;=\; \operatorname{Cov}(X_i, X_j) \;=\; \mathbb{E}[X_i X_j] \;-\; \mathbb{E}[X_i]\,\mathbb{E}[X_j].$$

The diagonal entries are variances, $\Sigma_{ii} = \operatorname{Var}(X_i)$, and the off-diagonals are pairwise covariances.

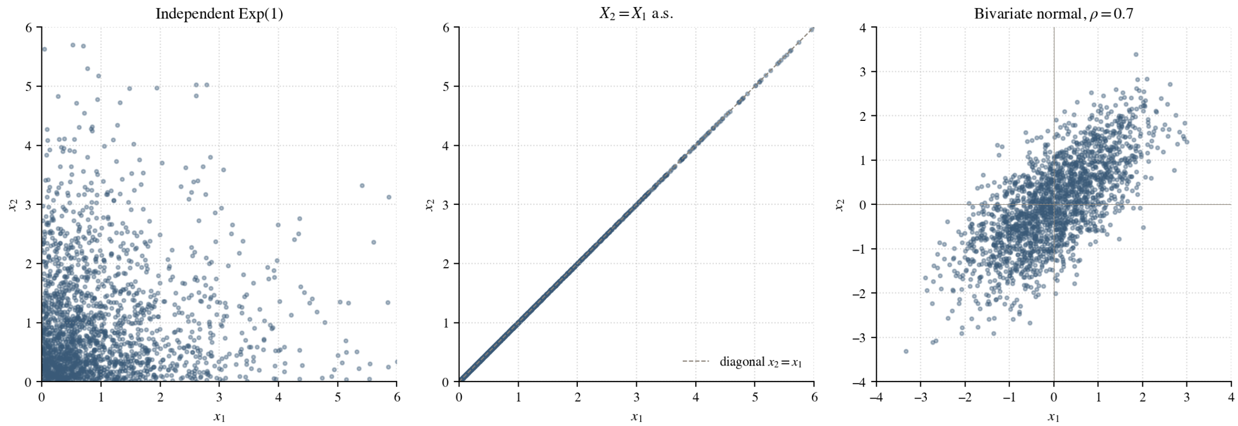

Figure 6: Three joint distributions on $\mathbb{R}^2$ with the same one-dimensional marginals. *Left:* two independent exponentials; the cloud fills the positive quadrant. *Middle:* $X_1 = X_2$ almost surely; the cloud collapses onto the diagonal line. *Right:* a bivariate normal with correlation $\rho = 0.7$; the cloud is an oblique elliptical scatter. Each panel shows 2,000 Monte Carlo samples. The middle case has no joint density on $\mathbb{R}^2$; it will be the prototype for a covariance matrix that is PSD but not PD.

The covariance matrix carries the first two moments of every linear combination of the coordinates of $\boldsymbol{X}$.

Proposition 2.13 (Σ is symmetric and PSD)**.** For any random vector with finite second moments:

(1) Σ is symmetric, $\Sigma_{ij} = \Sigma_{ji}$.

(2) Σ is positive semidefinite: $\boldsymbol{v}^\top \Sigma \boldsymbol{v} \geq 0$ for every $\boldsymbol{v} \in \mathbb{R}^p$. The identity $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = \operatorname{Var}(\boldsymbol{v}^\top \boldsymbol{X})$ shows that every quadratic form on Σ is a variance.

(3) Σ is positive definite if and only if no nontrivial linear combination $\boldsymbol{v}^\top \boldsymbol{X}$ ($\boldsymbol{v} \neq \boldsymbol{0}$) is almost surely constant.

Proof. Symmetry: $\Sigma_{ij} = \operatorname{Cov}(X_i, X_j) = \operatorname{Cov}(X_j, X_i) = \Sigma_{ji}$.

For (2), fix any $\boldsymbol{v} \in \mathbb{R}^p$ and define the scalar $Y = \boldsymbol{v}^\top \boldsymbol{X}$. Then

$$Y \; - \; \mathbb{E}[Y] \; = \; \boldsymbol{v}^\top \boldsymbol{X} \; - \; \boldsymbol{v}^\top \boldsymbol{\mu} \; = \; \boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu}),$$

and computing the variance gives

$$\operatorname{Var}(Y) \; = \; \mathbb{E}\big[(Y - \mathbb{E}[Y])^2\big] \; = \; \mathbb{E}\big[(\boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu}))^2\big].$$

Since $\boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu})$ is a scalar, its square equals $(\boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu}))(\boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu}))^\top = \boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top \boldsymbol{v}$. Pulling the constants $\boldsymbol{v}^\top$ and $\boldsymbol{v}$ outside the expectation,

$$\operatorname{Var}(Y) \; = \; \boldsymbol{v}^\top \, \mathbb{E}\big[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\big] \, \boldsymbol{v} \; = \; \boldsymbol{v}^\top \Sigma \boldsymbol{v}.$$

Since $\operatorname{Var}(Y) \geq 0$ for every random variable Y, we conclude $\boldsymbol{v}^\top \Sigma \boldsymbol{v} \geq 0$ for every $\boldsymbol{v}$, which is PSD.

For (3), Σ fails to be positive definite precisely when $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = 0$ for some $\boldsymbol{v} \neq \mathbf{0}$. By the identity above, this is $\mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{X}) = 0$, which is the statement that $\boldsymbol{v}^\top \boldsymbol{X}$ is almost surely a constant. $\square$

Example 2.14 is the prototypical failure of positive definiteness: with $X_2 = X_1$ a.s., the vector $\boldsymbol{v} = (1, -1)^\top$ gives $\boldsymbol{v}^\top \boldsymbol{X} = X_1 - X_2 = 0$ a.s., so $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = 0$. The covariance matrix is PSD but not PD.

The identity $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = \mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{X})$ is the structural fact that will organize the analysis of sample covariance matrices in Section 3: every spectral statement about Σ translates to a statement about variances of linear combinations of $\boldsymbol{X}$, and vice versa.

The multivariate normal distribution.

Definition 2.17 (Multivariate normal)**.** A random vector $\boldsymbol{X} \in \mathbb{R}^p$ is *multivariate normal* with mean $\boldsymbol{\mu}$ and covariance Σ, written $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, if its characteristic function is

$$\varphi_{\boldsymbol{X}}(\boldsymbol{t}) \;=\; \exp\Big(i\, \boldsymbol{t}^\top \boldsymbol{\mu} \;-\; \tfrac{1}{2}\, \boldsymbol{t}^\top \Sigma \boldsymbol{t}\Big), \qquad \boldsymbol{t} \in \mathbb{R}^p. \tag{4}$$

When Σ is positive definite, $\boldsymbol{X}$ admits the density

$$f_{\boldsymbol{X}}(\boldsymbol{x}) \;=\; \frac{1}{(2\pi)^{p/2}\, |\Sigma|^{1/2}} \exp\Big(-\tfrac{1}{2}\, (\boldsymbol{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu})\Big).$$

Equation (4) is taken as the definition because it remains valid when Σ is singular (a case that arises exactly as in Example 2.14), whereas the density form requires Σ to be invertible.

The multivariate normal is closed under affine transformations.

Proposition 2.14 (Affine closure of $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$)**.** If $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, $\boldsymbol{A} \in \mathbb{R}^{q \times p}$, and $\boldsymbol{b} \in \mathbb{R}^q$, then

$$\boldsymbol{A}\boldsymbol{X} + \boldsymbol{b} \;\sim\; \mathcal{N}(\boldsymbol{A}\boldsymbol{\mu} + \boldsymbol{b},\; \boldsymbol{A}\Sigma\boldsymbol{A}^\top).$$

Proof. We compute the characteristic function of $\boldsymbol{A}\boldsymbol{X} + \boldsymbol{b}$. For $\boldsymbol{s} \in \mathbb{R}^q$,

$$\begin{aligned}
\varphi_{\boldsymbol{A}\boldsymbol{X}+\boldsymbol{b}}(\boldsymbol{s}) \;&=\; \mathbb{E}\big[e^{i\boldsymbol{s}^\top(\boldsymbol{A}\boldsymbol{X}+\boldsymbol{b})}\big] \;=\; e^{i\boldsymbol{s}^\top \boldsymbol{b}}\, \mathbb{E}\big[e^{i(\boldsymbol{A}^\top \boldsymbol{s})^\top \boldsymbol{X}}\big] \\
&=\; e^{i\boldsymbol{s}^\top \boldsymbol{b}}\, \varphi_{\boldsymbol{X}}(\boldsymbol{A}^\top \boldsymbol{s}) \\
&=\; \exp\Big(i\, \boldsymbol{s}^\top(\boldsymbol{A}\boldsymbol{\mu} + \boldsymbol{b}) \;-\; \tfrac{1}{2}\, \boldsymbol{s}^\top \boldsymbol{A}\Sigma\boldsymbol{A}^\top \boldsymbol{s}\Big).
\end{aligned}$$

This is the CF of $\mathcal{N}(\boldsymbol{A}\boldsymbol{\mu} + \boldsymbol{b},\, \boldsymbol{A}\Sigma\boldsymbol{A}^\top)$, so the distribution of $\boldsymbol{A}\boldsymbol{X} + \boldsymbol{b}$ is identified by the uniqueness of characteristic functions. $\square$

A second algebraic property is special to Gaussians: uncorrelated jointly Gaussian variables are independent. For general distributions, $\mathrm{Cov}(X_1, X_2) = 0$ does not imply $X_1 \perp X_2$; for the multivariate normal it does, because the density factors as soon as Σ is diagonal (and the CF factors for the same reason in the singular case). Figure 7 shows the bivariate density contours at three correlation values.

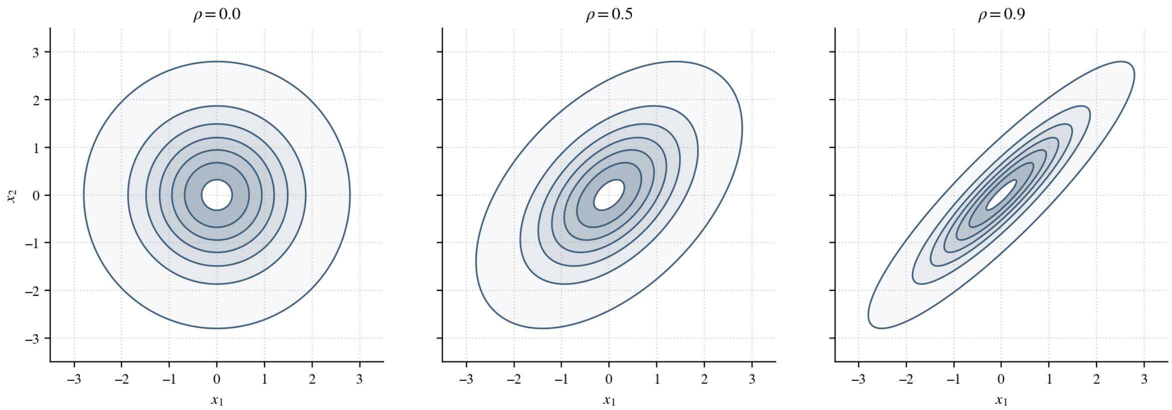

Figure 7: Density contours of the centered bivariate normal with unit variances and correlation $\rho \in \{0,\, 0.5,\, 0.9\}$. At $\rho = 0$ the contours are circles and the components are independent; as ρ increases the contours elongate along the line $x_2 = x_1$, reflecting positive linear association. Generated by `notebooks/01_lln_clt_demos.py`.

The multivariate central limit theorem. The multivariate CLT follows from the univariate version through the *Cramér-Wold device*: a sequence of random vectors converges in distribution if and only if every fixed linear combination of the coordinates converges in distribution. Applied to iid centered random vectors $\boldsymbol{X}_i$ with finite second moments and covariance Σ,

$$\sqrt{n}\,(\bar{\boldsymbol{X}}_n - \boldsymbol{\mu}) \;\xrightarrow{d}\; \mathcal{N}(\mathbf{0},\, \Sigma).$$

The vector CLT powers every fixed-p sample covariance calculation in Section 3 and sets up the contrast with the high-dimensional regime, where $\sqrt{n}$-Gaussian fluctuations no longer suffice to describe the spectral behavior of $\widehat{\Sigma}_n$.

2.7 Linear Algebra Preliminaries

The remainder of this section collects the linear-algebra results that the analysis of sample covariance matrices in Section 3 will rely on but that are not themselves probabilistic: the spectral theorem, the eigenvalue characterization of positive semidefiniteness, matrix norms, and the continuity of eigenvalues as functions of matrix entries. The probability content of Sections 2.1–2.6 gave us *what* sample covariance matrices are; the content here gives us the *tools* for analyzing them.

The spectral theorem. A real $n \times n$ matrix A is *symmetric* if $A = A^\top$. Throughout this paragraph, A denotes a real symmetric matrix. Symmetry has two crucial consequences: all eigenvalues of A are real, and there exists an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A.

Theorem 2.15 (Spectral theorem for real symmetric matrices)**.** If $A \in \mathbb{R}^{n\times n}$ is symmetric, there exist an orthogonal matrix Q (meaning $Q^\top Q = I$) and a diagonal matrix $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ such that

$$A \;=\; Q\,\Lambda\,Q^\top. \tag{5}$$

The diagonal entries of Λ are the eigenvalues of A, and the columns $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ of Q form an orthonormal basis of eigenvectors: $A\boldsymbol{q}_i = \lambda_i \boldsymbol{q}_i$ and $\boldsymbol{q}_i^\top \boldsymbol{q}_j = \delta_{ij}$.

A complete proof requires careful linear algebra; see Strang (2016, §6.4). We will use the conclusion freely. The decomposition $A = Q\Lambda Q^\top$ has a clean geometric meaning: the action of A on a vector $\boldsymbol{x} \in \mathbb{R}^n$ is (i) rotate $\boldsymbol{x}$ into the eigenvector basis via $Q^\top$, (ii) scale the i^{th} coordinate by λ_i via Λ, (iii) rotate back via Q. Symmetric matrices stretch space along perpendicular directions, with stretch factors given by the eigenvalues.

Quadratic forms diagonalize.

Corollary 2.16 (Diagonalization of quadratic forms)**.** For any $\boldsymbol{x} \in \mathbb{R}^n$, setting $\boldsymbol{y} = Q^\top \boldsymbol{x}$ gives

$$\boldsymbol{x}^\top A\,\boldsymbol{x} \;=\; \sum_{i=1}^{n} \lambda_i\, y_i^2.$$

Proof. Using $A = Q\Lambda Q^\top$,

$$\boldsymbol{x}^\top A\,\boldsymbol{x} \;=\; \boldsymbol{x}^\top Q\Lambda Q^\top \boldsymbol{x} \;=\; (Q^\top \boldsymbol{x})^\top\, \Lambda\,(Q^\top \boldsymbol{x}) \;=\; \boldsymbol{y}^\top \Lambda\, \boldsymbol{y}.$$

Since $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, the last expression is $\sum_{i=1}^n \lambda_i y_i^2$. □

Because Q is orthogonal, the map $\boldsymbol{x} \mapsto \boldsymbol{y} = Q^\top \boldsymbol{x}$ is a bijection of $\mathbb{R}^n$ that sends $\mathbf{0}$ to $\mathbf{0}$, so $\boldsymbol{x} \neq \mathbf{0}$ if and only if $\boldsymbol{y} \neq \mathbf{0}$. The sign of $\boldsymbol{x}^\top A\boldsymbol{x}$ is therefore controlled entirely by the signs of the eigenvalues.

Eigenvalues characterize positive definiteness.

Theorem 2.17 (Eigenvalue characterization of PD/PSD)**.** Let A be a real symmetric matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$.

(1) A is positive semidefinite if and only if $\lambda_i \geq 0$ for all i.

(2) A is positive definite if and only if $\lambda_i > 0$ for all i.

Proof. We prove (1); (2) is identical with strict inequalities throughout.

PSD $\Rightarrow \lambda_i \geq 0$. Suppose A is PSD, and let $\boldsymbol{q}_k$ be a unit eigenvector with eigenvalue λ_k. Computing the quadratic form at $\boldsymbol{q}_k$,

$$\boldsymbol{q}_k^\top A\,\boldsymbol{q}_k \;=\; \boldsymbol{q}_k^\top(\lambda_k \boldsymbol{q}_k) \;=\; \lambda_k\, \boldsymbol{q}_k^\top \boldsymbol{q}_k \;=\; \lambda_k.$$

By PSD the left side is ≥ 0, so $\lambda_k \geq 0$.

$\lambda_i \geq 0 \Rightarrow$ *PSD.* Suppose every eigenvalue of A is nonnegative. For any $\boldsymbol{x} \in \mathbb{R}^n$, Corollary 2.16 gives

$$\boldsymbol{x}^\top A\,\boldsymbol{x} \;=\; \sum_{i=1}^{n} \lambda_i\, y_i^2 \;\geq\; 0,$$

where $\boldsymbol{y} = Q^\top \boldsymbol{x}$ and each $\lambda_i y_i^2 \geq 0$ by hypothesis. Hence A is PSD. □

Combining Theorem 2.17 with Proposition 2.13, every eigenvalue of a covariance matrix is nonnegative, and the eigenvalues are strictly positive if and only if no nontrivial linear combination of the coordinates is almost surely constant. The remaining sample-covariance analysis in Section 3 will use this fact silently every time it manipulates an eigenvalue.

Worked examples in the 2×2 case. For a symmetric matrix $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, the characteristic polynomial $\det(A - \lambda I) = (a - \lambda)(d - \lambda) - b^2$ factors with eigenvalues

$$\lambda_{\pm} \;=\; \frac{a+d}{2} \;\pm\; \sqrt{\left(\frac{a-d}{2}\right)^2 + b^2}. \tag{6}$$

A convenient 2×2 test (a small case of Sylvester's criterion):

$$A \text{ is PD} \quad\iff\quad a > 0 \;\text{ and }\; \det A = ad - b^2 > 0.$$

Example 2.18 (A positive-definite and an indefinite 2×2)**.** *Positive-definite case.* Let $A_1 = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$. Equation (6) gives $\lambda_{\pm} = 1 \pm \frac{1}{2}$, so

$$\lambda_+ \;=\; \tfrac{3}{2}, \qquad \lambda_- \;=\; \tfrac{1}{2},$$

both strictly positive. Sylvester's criterion agrees: $a = 1 > 0$ and $\det A_1 = 1 - \frac{1}{4} = \frac{3}{4} > 0$. One can also exhibit the positivity directly by completing the square in the quadratic form:

$$\boldsymbol{x}^\top A_1 \boldsymbol{x} \;=\; x_1^2 - x_1 x_2 + x_2^2 \;=\; \left(x_1 - \tfrac{x_2}{2}\right)^2 + \tfrac{3}{4}\, x_2^2,$$

which is a sum of squares, manifestly > 0 for $\boldsymbol{x} \neq \mathbf{0}$.

Indefinite case. Let $A_2 = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}$. Equation (6) gives $\lambda_{\pm} = 1 \pm 2$, so $\lambda_+ = 3$ and $\lambda_- = -1$. The opposite signs make A_2 indefinite. Two directional certificates confirm this: $\boldsymbol{v} = (1,1)^\top$ gives $\boldsymbol{v}^\top A_2 \boldsymbol{v} = -2 < 0$, and $\boldsymbol{v} = (1,-1)^\top$ gives $\boldsymbol{v}^\top A_2 \boldsymbol{v} = 6 > 0$.

Figure 8 visualizes the geometry of the three regimes.

Vector and matrix norms. A *norm* on a real vector space V is a function $\|\cdot\| : V \to [0, \infty)$ satisfying, for all $\boldsymbol{v}, \boldsymbol{w} \in V$ and scalars α:

(N1) $\|\boldsymbol{v}\| = 0$ if and only if $\boldsymbol{v} = \mathbf{0}$ (definiteness);

(N2) $\|\alpha \boldsymbol{v}\| = |\alpha|\, \|\boldsymbol{v}\|$ (homogeneity);

(N3) $\|\boldsymbol{v} + \boldsymbol{w}\| \le \|\boldsymbol{v}\| + \|\boldsymbol{w}\|$ (triangle inequality).

On $\mathbb{R}^p$ the standard examples are

$$\|\boldsymbol{v}\|_2 \;=\; \Big(\sum_{i=1}^p v_i^2\Big)^{1/2}, \qquad \|\boldsymbol{v}\|_1 \;=\; \sum_{i=1}^p |v_i|, \qquad \|\boldsymbol{v}\|_\infty \;=\; \max_i |v_i|;$$

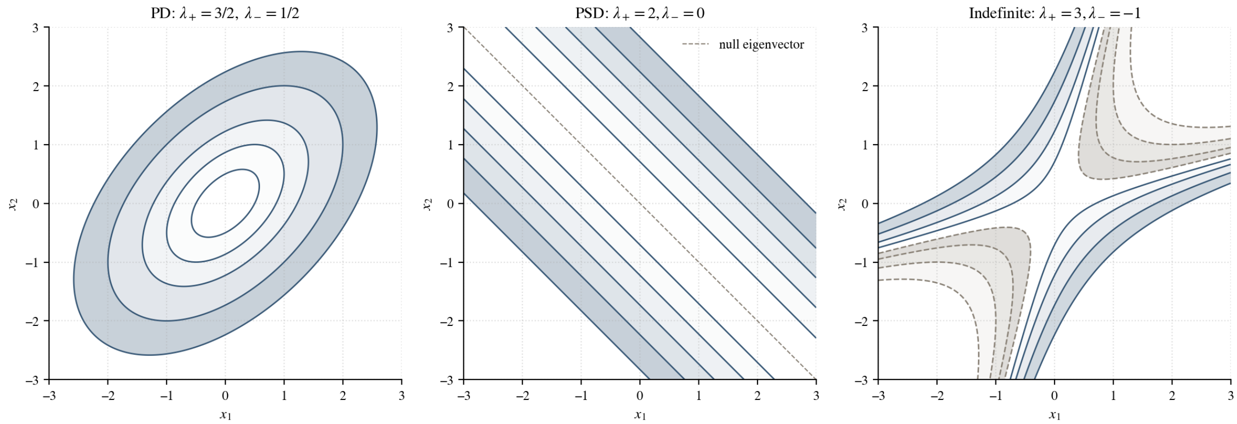

Figure 8: Level sets of $\boldsymbol{x}^\top A\,\boldsymbol{x}$ for three real symmetric 2×2 matrices, illustrating Theorem 2.17. *Left:* a positive-definite A with both eigenvalues positive; level sets are nested ellipses with axes aligned with the eigenvectors. *Middle:* a positive-semidefinite A with one zero eigenvalue; level sets degenerate into parallel strips along the null eigenvector. *Right:* an indefinite A with eigenvalues of opposite sign; level sets are hyperbolas and the quadratic form is positive in some directions and negative in others.

each satisfies (N1)–(N3) by direct verification (the triangle inequality for $\|\cdot\|_2$ is the Cauchy-Schwarz inequality applied to $\|\boldsymbol{v}+\boldsymbol{w}\|_2^2$).

On $\mathbb{R}^{p\times p}$ we use two matrix norms.

Definition 2.19 (Frobenius and operator norms)**.** For $A \in \mathbb{R}^{p\times p}$,

$$\|A\|_F \;=\; \sqrt{\sum_{i=1}^{p}\sum_{j=1}^{p} a_{ij}^2}, \qquad \|A\|_{\mathrm{op}} \;=\; \sup_{\substack{\boldsymbol{x}\in\mathbb{R}^p\\ \|\boldsymbol{x}\|_2=1}} \|A\boldsymbol{x}\|_2.$$

The Frobenius norm is the Euclidean norm of A viewed as a vector in $\mathbb{R}^{p^2}$; it satisfies (N1)–(N3) for this reason, and

$$A_n \to A \text{ entrywise} \quad\iff\quad \|A_n - A\|_F \to 0.$$

The operator norm measures the largest factor by which A stretches a unit vector; for symmetric A it equals the largest absolute eigenvalue, $\|A\|_{\mathrm{op}} = \max_i |\lambda_i|$.

Theorem 2.18 (Equivalence of norms in finite dimensions)**.** For any two norms $\|\cdot\|_a, \|\cdot\|_b$ on a finite-dimensional real vector space, there exist constants $c, C > 0$ such that

$$c\,\|\boldsymbol{v}\|_a \;\le\; \|\boldsymbol{v}\|_b \;\le\; C\,\|\boldsymbol{v}\|_a \qquad \text{for every } \boldsymbol{v}.$$

The proof is a compactness argument: the function $\boldsymbol{v} \mapsto \|\boldsymbol{v}\|_b/\|\boldsymbol{v}\|_a$ is continuous and positive on the unit sphere $\{\|\boldsymbol{v}\|_a = 1\}$, which is compact in finite dimensions; its minimum c and maximum C are strictly positive, and the inequality extends to all $\boldsymbol{v}$ by homogeneity. The practical consequence for our purposes is that for matrices in finite dimensions, every sense of convergence to zero (entrywise, Frobenius, operator) implies all of the others.

Determinant and the characteristic polynomial. The Leibniz formula expresses the determinant as a polynomial in the matrix entries:

$$\det A \;=\; \sum_{\pi \in S_p} \operatorname{sgn}(\pi) \prod_{i=1}^{p} a_{i,\pi(i)}, \tag{7}$$

where S_p is the group of permutations of $\{1, \ldots, p\}$. Since polynomials in the entries are continuous functions of the entries, so is the determinant:

$$A_n \to A \text{ entrywise} \quad \Longrightarrow \quad \det A_n \to \det A.$$

The *characteristic polynomial* of A is

$$p_A(\lambda) \;=\; \det(A - \lambda I), \qquad \lambda \in \mathbb{R},$$

whose roots are exactly the eigenvalues of A, counted with algebraic multiplicity. The coefficients of p_A are themselves polynomials in the entries of A (apply Leibniz to $A - \lambda I$), so they too are continuous functions of A.

Eigenvalue convergence. We can now state the key linear-algebra ingredient powering Section 3: when a sequence of matrices converges, their eigenvalues converge. The deterministic statement is a standard fact about polynomial roots; we then push it through the modes of convergence developed in Section 2.4, following Genzer (2025, Prop. 3.1).

Proposition 2.19 (Eigenvalue convergence via characteristic polynomials)**.** Let $\{\Sigma_n\}_{n\geq 1}$ and Σ_0 be $p \times p$ symmetric matrices, and write $\lambda_i(\Sigma)$ for the i^{th} eigenvalue of Σ (with some fixed ordering convention).

(1) *Deterministic.* If $\Sigma_n \to \Sigma_0$ entrywise, then $\lambda_i(\Sigma_n) \to \lambda_i(\Sigma_0)$ for every i.

(2) *Almost sure.* If $\Sigma_n \xrightarrow{\text{a.s.}} \Sigma_0$ in any matrix norm, then $\lambda_i(\Sigma_n) \xrightarrow{\text{a.s.}} \lambda_i(\Sigma_0)$.

(3) *In probability.* If $\Sigma_n \xrightarrow{\mathbb{P}} \Sigma_0$ in any matrix norm, then $\lambda_i(\Sigma_n) \xrightarrow{\mathbb{P}} \lambda_i(\Sigma_0)$.

Proof. Step 1 (deterministic). Suppose $\Sigma_n \to \Sigma_0$ entrywise. By the Leibniz formula (7), the coefficients of the characteristic polynomial $p_{\Sigma_n}(\lambda) = \det(\Sigma_n - \lambda I)$ are polynomial functions of the entries of Σ_n, so they converge to the coefficients of p_{Σ_0}. A standard result from analysis states that the roots of a monic polynomial depend continuously on its coefficients (after a consistent ordering); applied to the characteristic polynomials this gives $\lambda_i(\Sigma_n) \to \lambda_i(\Sigma_0)$ for every i.

Step 2 (almost sure). Suppose $\Sigma_n \xrightarrow{\text{a.s.}} \Sigma_0$ in some matrix norm $\|\cdot\|$. By Theorem 2.18, this is equivalent to convergence in Frobenius norm, which in turn is entrywise convergence (with probability one). Concretely, there is an event Ω_0 with $\mathbb{P}(\Omega_0) = 1$ such that for every $\omega \in \Omega_0$, $\Sigma_n(\omega) \to \Sigma_0(\omega)$ entrywise. Step 1 applies to this deterministic sequence and yields $\lambda_i(\Sigma_n(\omega)) \to \lambda_i(\Sigma_0(\omega))$ for every $\omega \in \Omega_0$, which is the statement $\lambda_i(\Sigma_n) \xrightarrow{\text{a.s.}} \lambda_i(\Sigma_0)$.

Step 3 (in probability). Suppose $\Sigma_n \xrightarrow{\mathbb{P}} \Sigma_0$ in some matrix norm. The map $\Sigma \mapsto (\lambda_1(\Sigma), \ldots, \lambda_p(\Sigma))$ is continuous (entries to coefficients to roots), so for every $\eta > 0$ there exists $\varepsilon > 0$ such that

$$\|\Sigma - \Sigma_0\| \; < \; \varepsilon \quad \Longrightarrow \quad \max_i |\lambda_i(\Sigma) - \lambda_i(\Sigma_0)| \; < \; \eta.$$

Translating to events,

$$\Big\{\max_i |\lambda_i(\Sigma_n) - \lambda_i(\Sigma_0)| \geq \eta\Big\} \; \subseteq \; \big\{\|\Sigma_n - \Sigma_0\| \geq \varepsilon\big\},$$

and taking probabilities,

$$\mathbb{P}\Big(\max_i |\lambda_i(\Sigma_n) - \lambda_i(\Sigma_0)| \geq \eta\Big) \; \leq \; \mathbb{P}(\|\Sigma_n - \Sigma_0\| \geq \varepsilon) \; \longrightarrow \; 0.$$

This is $\lambda_i(\Sigma_n) \xrightarrow{\mathbb{P}} \lambda_i(\Sigma_0)$ for every i. $\square$

The deterministic chain *entries* $\to$ *characteristic-polynomial coefficients* $\to$ *roots* is the entire logical content. The almost-sure and in-probability versions follow by pushing continuity through the respective modes of convergence. In Section 3 this proposition will deliver, with no further work, that the eigenvalues of the sample covariance matrix $\widehat{\Sigma}_n$ converge to those of the population covariance Σ whenever $\widehat{\Sigma}_n$ converges to Σ in any of the standard senses.

3 Sample Covariance Matrices

Section 2.6 introduced the covariance matrix Σ of a random vector as a theoretical object with entries $\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$. In practice this matrix is unknown; what one has in hand is a sample $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ and an estimator built from it. The natural estimator is the *sample covariance matrix* $\widehat{\Sigma}_n$, the central object of this report.

The remainder of this section asks two questions about $\widehat{\Sigma}_n$. First, in what sense does $\widehat{\Sigma}_n \to \Sigma$ when p is fixed and $n \to \infty$, and how do its eigenvalues and eigenvectors compare to those of Σ? Second, what happens when p and n grow together with $p/n \to y > 0$? The answers, given in Sections 3.2 and 3.3, are qualitatively different, and that contrast is the central scientific point of this report.

3.1 The Sample Covariance Matrix

For random vectors $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ with mean $\boldsymbol{\mu}$ and population covariance Σ, the most concrete way to define an estimator of Σ is entrywise. The (j,k) entry of Σ is $\Sigma_{jk} = \mathbb{E}[X_{1j}X_{1k}] - \mu_j\mu_k$; we estimate it by the corresponding sample average. Specializing to the mean-zero case $\boldsymbol{\mu} = \mathbf{0}$ (used throughout the rest of this report), the natural estimator is

$$(\widehat{\Sigma}_n)_{jk} \;=\; \frac{1}{n}\sum_{i=1}^{n} X_{ij}X_{ik}, \qquad j, k = 1, \ldots, p.$$

The diagonal entries $(\widehat{\Sigma}_n)_{jj} = n^{-1}\sum_i X_{ij}^2$ are sample variances, and the off-diagonal entries are sample covariances; both are sample averages of products of coordinate values across the n observations.

Definition 3.1 (Sample covariance matrix)**.** Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ be iid copies of a random vector $\boldsymbol{X} \in \mathbb{R}^p$. In the *mean-zero* case ($\boldsymbol{\mu} = \mathbf{0}$ known), the *sample covariance matrix* is the $p \times p$ matrix

$$\widehat{\Sigma}_n \;=\; \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{X}_i\boldsymbol{X}_i^\top, \tag{8}$$

with entries $(\widehat{\Sigma}_n)_{jk} = n^{-1}\sum_i X_{ij}X_{ik}$. In the *mean-unknown* case, the (unbiased) sample covariance matrix is

$$\widehat{\Sigma}_n \;=\; \frac{1}{n-1}\sum_{i=1}^{n}(\boldsymbol{X}_i - \bar{\boldsymbol{X}}_n)(\boldsymbol{X}_i - \bar{\boldsymbol{X}}_n)^\top, \qquad \bar{\boldsymbol{X}}_n \;=\; \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{X}_i.$$

We work in the mean-zero case throughout the report; the Gaussian models that drive Sections 3 and 4 all have $\boldsymbol{\mu} = \mathbf{0}$ by convention, and the matrix algebra is cleaner. The asymptotic results below carry over to the mean-unknown version with no substantive change.

Matrix form. Stacking the n observations as the columns of the *data matrix*

$$\boldsymbol{X} \;=\; (\boldsymbol{X}_1 \mid \cdots \mid \boldsymbol{X}_n) \;\in\; \mathbb{R}^{p\times n}$$

gives a compact rewriting of equation (8): the sum of outer products is exactly $\boldsymbol{X}\boldsymbol{X}^\top$, so

$$\widehat{\Sigma}_n = \frac{1}{n}\boldsymbol{X}\boldsymbol{X}^\top. \tag{9}$$

A direct check confirms that (9) agrees entrywise with (8): the (j,k) entry of $\boldsymbol{X}\boldsymbol{X}^\top$ is $\sum_i(\boldsymbol{X})_{ji}(\boldsymbol{X}^\top)_{ik} = \sum_i X_{ij}X_{ik}$. The matrix form is the version we will manipulate throughout; up to a factor of n, it is the Gram matrix of the data.

Geometric meaning. The eigenvectors of Σ are the directions of greatest, second-greatest, and so on, variance of $\boldsymbol{X}$; the eigenvalues record the variance along each direction. The eigenvectors of $\widehat{\Sigma}_n$ are the sample analogues. Figure 9 shows the picture in two dimensions: for a sample drawn from $\mathcal{N}(\mathbf{0}, \Sigma)$, the sample principal axes track the population principal axes closely, and the alignment is exact in the limit $n \to \infty$ (we make this precise in Theorem 3.2 below).

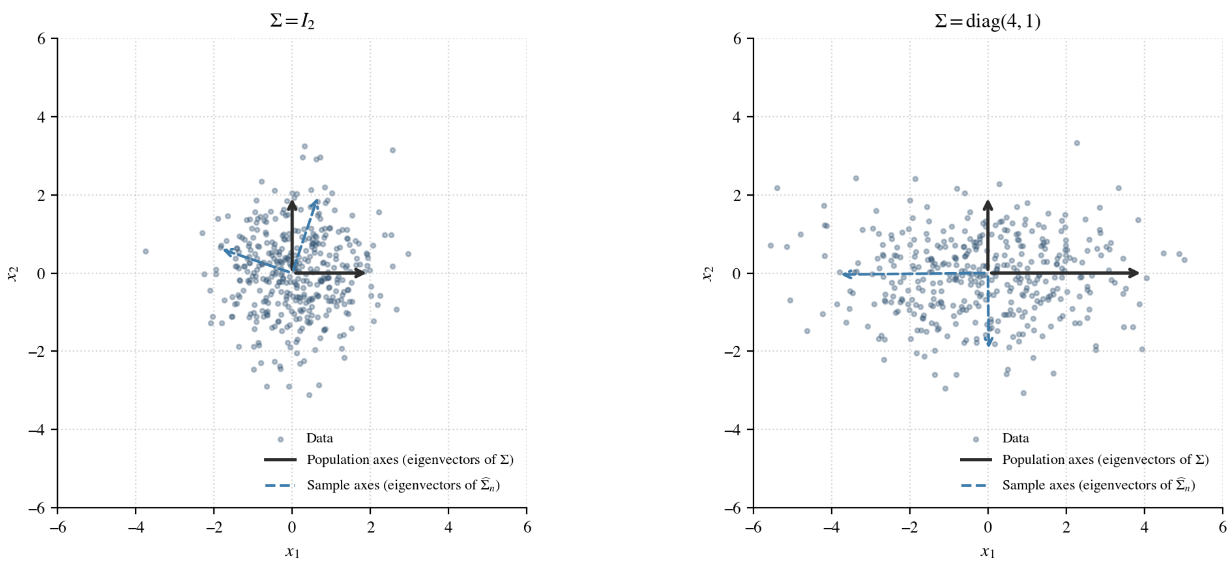

Figure 9: Sample covariance captures the population principal axes. Each panel shows $n = 400$ iid samples from $\mathcal{N}(\mathbf{0}, \Sigma)$ in two dimensions, with the eigenvectors of Σ (solid arrows, scaled by $\sqrt{\lambda_i(\Sigma)}$) and the eigenvectors of $\widehat{\Sigma}_n$ (dashed arrows, scaled by $\sqrt{\lambda_i(\widehat{\Sigma}_n)}$) overlaid. *Left:* $\Sigma = I_2$; the cloud is circular and the population axes are arbitrary, but the sample axes still pick out two perpendicular directions of comparable length. *Right:* $\Sigma = \mathrm{diag}(4, 1)$; the cloud is elongated along the x_1 axis, and both the population and sample axes track the elongation. The sample axes deviate slightly from the population axes due to finite-sample noise.

A small worked example. The following two-dimensional example, drawn from a hypothetical pair of exam scores, illustrates how Definition 3.1 and equation (9) translate into a concrete computation. (We use the mean-unknown version here because the means are not known to be zero.)

Example 3.2 (Sample covariance from three two-dimensional observations)**.** Three observations on two variables are

$$\boldsymbol{x}_1 = (70, 75)^\top, \quad \boldsymbol{x}_2 = (80, 85)^\top, \quad \boldsymbol{x}_3 = (90, 80)^\top.$$

Step one: compute the sample means. They are $\bar{x}_1 = 80$ and $\bar{x}_2 = 80$, so the centered observations are

$$\boldsymbol{x}_1 - \bar{\boldsymbol{x}} = (-10, -5)^\top, \quad \boldsymbol{x}_2 - \bar{\boldsymbol{x}} = (0, 5)^\top, \quad \boldsymbol{x}_3 - \bar{\boldsymbol{x}} = (10, 0)^\top.$$

Step two: sum the outer products. The three outer products are

$$\begin{pmatrix} 100 & 50 \\ 50 & 25 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 25 \end{pmatrix}, \quad \begin{pmatrix} 100 & 0 \\ 0 & 0 \end{pmatrix},$$

summing to

$$\sum_{i=1}^{3} (\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^\top \;=\; \begin{pmatrix} 200 & 50 \\ 50 & 50 \end{pmatrix}.$$

Step three: divide by $n - 1 = 2$ to obtain

$$\widehat{\Sigma}_3 \;=\; \begin{pmatrix} 100 & 25 \\ 25 & 25 \end{pmatrix}.$$

The diagonal entries (100 and 25) are sample variances; the off-diagonal entry (25) is the sample covariance, indicating positive linear association between the two variables.

Basic structural properties. The sample covariance matrix inherits the symmetry and positive semidefiniteness of the population covariance.

Proposition 3.1 ($\widehat{\Sigma}_n$ is symmetric and PSD)**.** The sample covariance matrix $\widehat{\Sigma}_n$ is symmetric. It is positive semidefinite, so by Theorem 2.17 all of its eigenvalues are nonnegative.

Proof. For the mean-zero version (9), symmetry is immediate: $(\boldsymbol{X}\boldsymbol{X}^\top)^\top = (\boldsymbol{X}^\top)^\top \boldsymbol{X}^\top = \boldsymbol{X}\boldsymbol{X}^\top$. For positive semidefiniteness, fix any $\boldsymbol{v} \in \mathbb{R}^p$ and write

$$\boldsymbol{v}^\top \widehat{\Sigma}_n \boldsymbol{v} \;=\; \frac{1}{n} \boldsymbol{v}^\top \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{v} \;=\; \frac{1}{n} (\boldsymbol{X}^\top \boldsymbol{v})^\top (\boldsymbol{X}^\top \boldsymbol{v}) \;=\; \frac{1}{n} \|\boldsymbol{X}^\top \boldsymbol{v}\|_2^2 \;\geq\; 0.$$

The eigenvalue conclusion follows from Theorem 2.17. The mean-unknown version uses the centered data matrix in place of $\boldsymbol{X}$; the same identity gives $\boldsymbol{v}^\top \widehat{\Sigma}_n \boldsymbol{v} = (n-1)^{-1} \|\boldsymbol{X}_c^\top \boldsymbol{v}\|_2^2 \geq 0$. □

The remainder of this section divides along the two regimes set up at the start: low-dimensional in Section 3.2 (where p is fixed and n grows), and high-dimensional in Section 3.3 (where p grows with n).

3.2 Low Dimensions: Consistency, Wishart, and Fluctuations

In the low-dimensional regime the dimension p is fixed and the sample size $n \to \infty$. This is the classical setting of multivariate statistics, in which the limit theorems of Section 2 apply directly to the entries of $\widehat{\Sigma}_n$. We unpack four nested questions: (i) does $\widehat{\Sigma}_n$ converge to Σ, (ii) what is the exact distribution of $\widehat{\Sigma}_n$ for Gaussian data, (iii) what is the limiting distribution of the fluctuation $\sqrt{n}(\widehat{\Sigma}_n - \Sigma)$, and (iv) what does that say about individual eigenvalues and eigenvectors of $\widehat{\Sigma}_n$?

Consistency. The first question has the cleanest answer. Figure 10 demonstrates it numerically: the Frobenius error $\|\widehat{\Sigma}_n - \Sigma\|_F$ decays like $n^{-1/2}$ for any fixed p, and goes to zero with high probability as n grows. The exact statement and proof are immediate from the strong law of large numbers in Section 2.5.

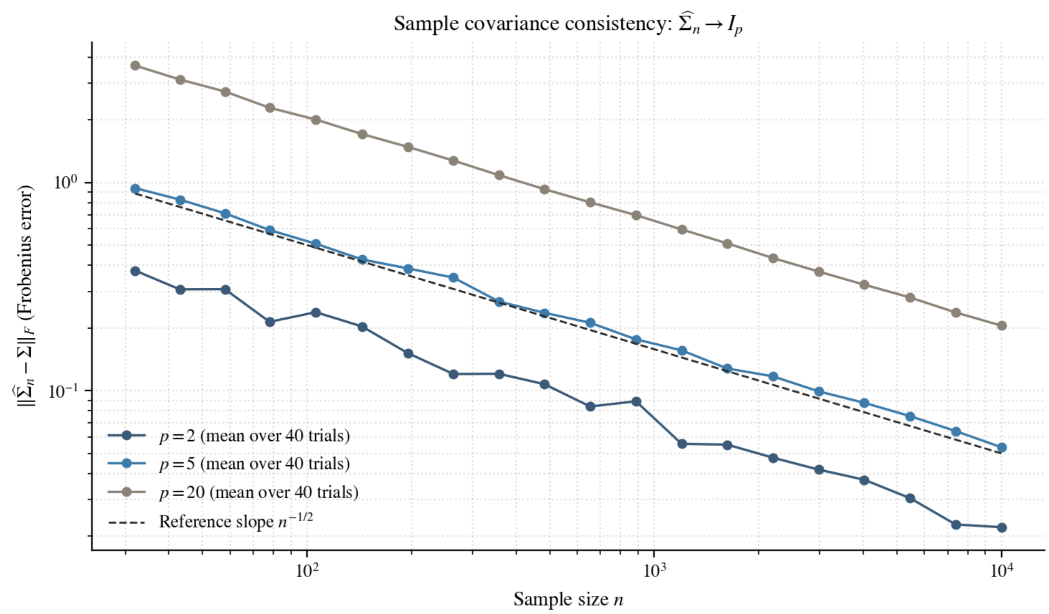

Figure 10: Sample covariance consistency in the low-dimensional regime. For $\Sigma = I_p$ and iid Gaussian samples, the Frobenius error $\|\widehat{\Sigma}_n - \Sigma\|_F$ averaged over 40 Monte Carlo trials is plotted against n (log-log axes) for three values of p. All three curves are parallel to the reference line of slope $n^{-1/2}$ (dashed), confirming that the error vanishes at the expected rate. Theorem 3.2 below proves the entrywise version of this convergence.

Theorem 3.2 (Entrywise consistency of $\widehat{\Sigma}_n$)**.** Let $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots$ be iid copies of $\boldsymbol{X} \in \mathbb{R}^p$ with mean $\mathbf{0}$ and finite second moments $\mathbb{E}[X_j X_k] < \infty$. Then

$$\widehat{\Sigma}_n \xrightarrow{\text{a.s.}} \Sigma \qquad \text{entrywise, as } n \to \infty.$$

Proof. The (j, k) entry of $\widehat{\Sigma}_n$ is

$$(\widehat{\Sigma}_n)_{jk} \;=\; \frac{1}{n} \sum_{i=1}^{n} X_{ij}\, X_{ik}.$$

The random variables $\{X_{ij}X_{ik}\}_{i\geq 1}$ are iid with mean

$$\mathbb{E}[X_{1j}X_{1k}] \;=\; \Sigma_{jk}$$

(using $\boldsymbol{\mu} = \mathbf{0}$) and finite first moment by hypothesis. Kolmogorov's strong law of large numbers (Theorem 2.10) applied to this sequence yields

$$(\widehat{\Sigma}_n)_{jk} \;\xrightarrow{\text{a.s.}}\; \Sigma_{jk}.$$

Doing this for each pair (j,k) gives entrywise almost-sure convergence $\widehat{\Sigma}_n \xrightarrow{\text{a.s.}} \Sigma$. □

Combined with the eigenvalue-convergence machinery developed in Section 2.7, entrywise convergence forces convergence of the eigenvalues themselves.

Corollary 3.3 (Sample eigenvalues converge to population eigenvalues)**.** Under the hypotheses of Theorem 3.2, $\lambda_i(\widehat{\Sigma}_n) \xrightarrow{\text{a.s.}} \lambda_i(\Sigma)$ for every $i = 1, \ldots, p$.

Proof. Apply Proposition 2.19(2) (the almost-sure version of eigenvalue convergence) to the random matrix sequence $\widehat{\Sigma}_n \xrightarrow{\text{a.s.}} \Sigma$. □

Theorem 3.2 and Corollary 3.3 together give the qualitative low-dimensional picture: $\widehat{\Sigma}_n$ is a consistent estimator of Σ, and the sample eigenvalues track the population eigenvalues. The remaining questions concern the size and shape of the fluctuations.

The Wishart distribution. When the observations are Gaussian, $\widehat{\Sigma}_n$ (more precisely, $n\widehat{\Sigma}_n$) has an exact distribution with a name. This makes it a tractable starting point for finer questions.

Definition 3.3 (Wishart distribution)**.** Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ be iid $\mathcal{N}(\mathbf{0}, \Sigma)$ random vectors in $\mathbb{R}^p$, and set

$$\boldsymbol{S} \;=\; \sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{X}_i^\top \;=\; n\,\widehat{\Sigma}_n.$$

The distribution of $\boldsymbol{S}$ is the *Wishart distribution* with n degrees of freedom and scale matrix Σ, written $\boldsymbol{S} \sim W_p(n, \Sigma)$. When $n \geq p$ and Σ is positive definite, $\boldsymbol{S}$ admits the density (on the cone of positive-definite matrices)

$$f_{\boldsymbol{S}}(\boldsymbol{S}) \;=\; \frac{|\boldsymbol{S}|^{(n-p-1)/2}\,\exp\!\left(-\frac{1}{2}\,\operatorname{tr}(\Sigma^{-1}\boldsymbol{S})\right)}{2^{np/2}\,|\Sigma|^{n/2}\,\Gamma_p(n/2)}, \tag{10}$$

where Γ_p is the multivariate gamma function; see Yao et al. (2015, §3.2) for a derivation.

The Wishart distribution is the multivariate analogue of the chi-square distribution, and recovers it in dimension one.

Example 3.4 (Wishart in dimension one)**.** Take $p = 1$ and $\Sigma = 1$. Then each $\boldsymbol{X}_i$ is a scalar $X_i \sim \mathcal{N}(0, 1)$, so $\boldsymbol{S} = \sum_{i=1}^n X_i^2$. By the definition of the chi-square distribution, $\boldsymbol{S} \sim \chi_n^2$. Substituting $p = 1$, $\Sigma = 1$, and $\mathrm{tr}(\Sigma^{-1}\boldsymbol{S}) = \boldsymbol{S}$ into (10) confirms this:

$$f_{\boldsymbol{S}}(s) \;=\; \frac{s^{(n-2)/2}\, e^{-s/2}}{2^{n/2}\,\Gamma(n/2)}, \qquad s > 0,$$

which is exactly the χ_n^2 density.

Eigenvalue repulsion in two dimensions. For $\Sigma = I_2$, the joint distribution of the two ordered eigenvalues $0 \le \lambda_1 \le \lambda_2$ of $\boldsymbol{S} \sim W_2(n, I_2)$ admits an explicit density,

$$f(\lambda_1, \lambda_2) \;=\; C\,(\lambda_1\lambda_2)^{(n-3)/2}\, e^{-(\lambda_1+\lambda_2)/2}\,(\lambda_2 - \lambda_1)\,\mathbf{1}_{\{0\le\lambda_1\le\lambda_2\}}, \tag{11}$$

where $C > 0$ is a normalizing constant; a derivation via a Jacobian computation on the spectral decomposition $\boldsymbol{S} = \boldsymbol{Q}\,\mathrm{diag}(\lambda_1, \lambda_2)\boldsymbol{Q}^\top$ is given in Yao et al. (2015, Ch. 3). The key feature of (11) is the factor $(\lambda_2 - \lambda_1)$, which forces the joint density to vanish on the diagonal $\lambda_1 = \lambda_2$. This is the simplest case of *eigenvalue repulsion*: the two eigenvalues of a random symmetric matrix avoid each other. Figure 11 visualizes the theoretical density, and Figure 12 confirms repulsion in simulation.

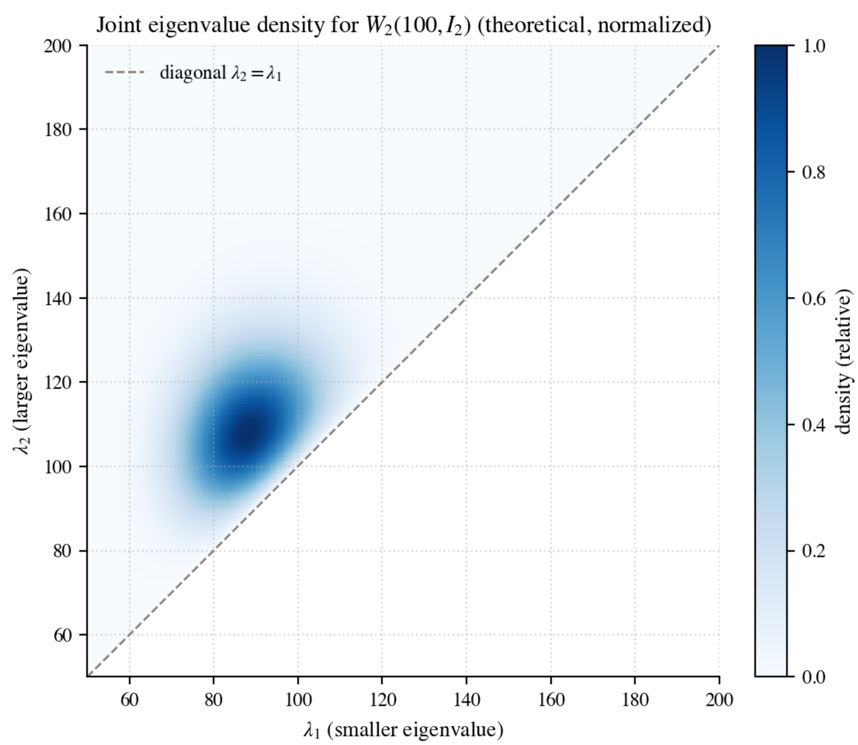

Figure 11: The theoretical joint eigenvalue density (11) of $W_2(100, I_2)$ on the half-plane $\lambda_1 \le \lambda_2$. The density vanishes along the diagonal (dashed line) because of the $(\lambda_2 - \lambda_1)$ factor; this dead zone is the analytic source of eigenvalue repulsion.

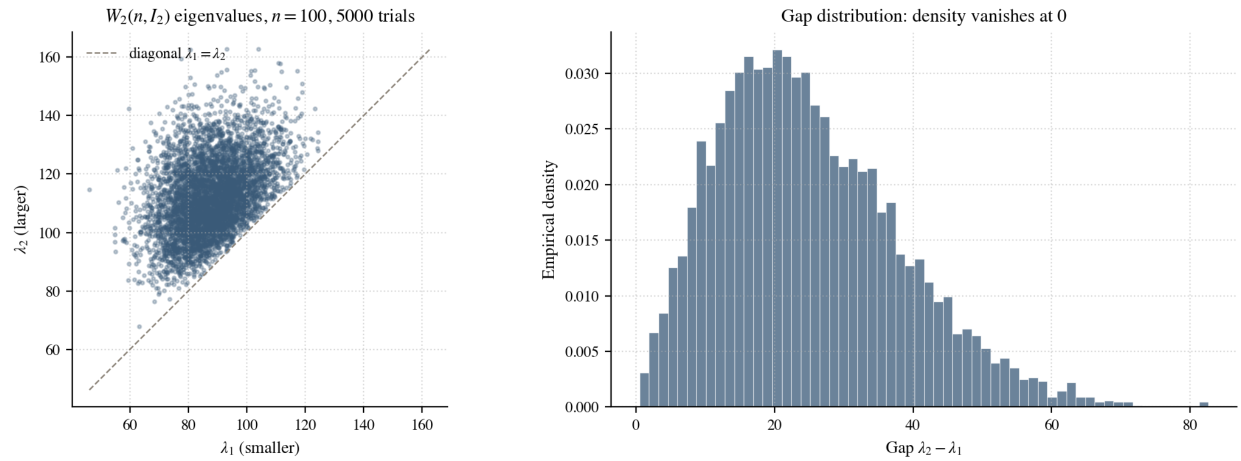

Figure 12: Eigenvalues of $W_2(100, I_2)$ computed from 5000 independent Monte Carlo trials. *Left:* scatter plot of (λ_1, λ_2) with the line $\lambda_2 = \lambda_1$ shown for reference. The cloud is visibly bounded away from the diagonal, mirroring the dead zone in Figure 11. *Right:* histogram of the gap $\lambda_2 - \lambda_1$; the density vanishes at zero, the empirical signature of repulsion. Generated by `notebooks/03_section3_demos.py`.

The Gaussian Orthogonal Ensemble. The eigenvalue repulsion that appears in the Wishart density (11) also appears, with the same $(\lambda_2 - \lambda_1)$ factor, in a different random matrix model: the Gaussian Orthogonal Ensemble. We introduce it here because it is the limit object in the matrix-valued CLT below.

Definition 3.5 (Gaussian Orthogonal Ensemble, 2×2 case)**.** A 2×2 random matrix

$$\boldsymbol{G} = \begin{pmatrix} G_{11} & G_{12} \\ G_{12} & G_{22} \end{pmatrix}$$

is said to belong to the *Gaussian Orthogonal Ensemble* (GOE_2) if its three independent entries are Gaussian with

$$G_{11},\, G_{22} \sim \mathcal{N}(0,\, 2), \qquad G_{12} \sim \mathcal{N}(0,\, 1),$$

and the three entries are mutually independent. More generally, a $p \times p$ GOE matrix has independent Gaussian entries on and above the diagonal with diagonal variance 2 and off-diagonal variance 1.

The variances 2 on the diagonal and 1 off the diagonal are chosen exactly so that the matrix is invariant in distribution under orthogonal conjugation $\boldsymbol{G} \mapsto \boldsymbol{O}\boldsymbol{G}\boldsymbol{O}^\top$; this is the eponymous orthogonal invariance. The top eigenvalue of GOE_2 is a non-Gaussian random variable, with a density slightly right-skewed by the repulsion that mirrors (11).

The matrix-valued central limit theorem. The next step transports the multivariate CLT of Section 2.6 to a statement about the entire fluctuation matrix $\sqrt{n}(\widehat{\Sigma}_n - \Sigma)$. For the null case $\Sigma = I_2$, the limit is exactly the GOE.

Proposition 3.4 (Matrix-valued CLT for $\Sigma = I_2$)**.** Let $\boldsymbol{X}_i \sim \mathcal{N}(\mathbf{0}, I_2)$ be iid. Then $\sqrt{n}\,(\widehat{\Sigma}_n - I_2)$ converges in distribution, as $n \to \infty$, to a GOE_2 matrix.

Proof. The three independent entries of the symmetric matrix $\sqrt{n}\,(\widehat{\Sigma}_n - I_2)$ are

$$\sqrt{n}\,((\widehat{\Sigma}_n)_{11} - 1), \quad \sqrt{n}\,((\widehat{\Sigma}_n)_{22} - 1), \quad \sqrt{n}\,(\widehat{\Sigma}_n)_{12},$$

each of which is a sum of iid centered random variables, so the multivariate central limit theorem of Section 2.6 applies. We compute the limiting variances and verify that the cross-covariances vanish.

For the diagonal entries, write $(\widehat{\Sigma}_n)_{jj} - 1 = n^{-1}\sum_i (X_{ij}^2 - 1)$. Each summand has mean zero and

$$\mathrm{Var}(X_{ij}^2) \;=\; \mathbb{E}[X_{ij}^4] \;-\; (\mathbb{E}[X_{ij}^2])^2 \;=\; 3 \;-\; 1 \;=\; 2,$$

using $\mathbb{E}[Z^4] = 3$ for $Z \sim \mathcal{N}(0,1)$. Hence the limiting variance of $\sqrt{n}\,((\widehat{\Sigma}_n)_{jj} - 1)$ is 2.

For the off-diagonal entry, write $(\widehat{\Sigma}_n)_{12} = n^{-1}\sum_i X_{i1}X_{i2}$. Each summand has mean zero (independence) and variance

$$\mathrm{Var}(X_{i1}X_{i2}) \;=\; \mathbb{E}[X_{i1}^2]\,\mathbb{E}[X_{i2}^2] \;=\; 1,$$

again by independence. Hence the limiting variance of $\sqrt{n}\,(\widehat{\Sigma}_n)_{12}$ is 1.

The cross-covariances vanish:

$$\mathrm{Cov}(X_{i1}^2,\, X_{i1}X_{i2}) \;=\; \mathbb{E}[X_{i1}^3]\,\mathbb{E}[X_{i2}] \;=\; 0,$$

because $\mathbb{E}[X_{i2}] = 0$, and analogously for the other pairs.

The three limiting variances are therefore 2, 2, 1 and the three limits are uncorrelated. By the multivariate CLT, the joint limit is $\mathcal{N}(\mathbf{0},\, \mathrm{diag}(2,2,1))$, which is exactly the entrywise distribution of a GOE_2 matrix from Definition 3.5. $\square$

First-order eigenvalue perturbation. Proposition 3.4 describes the fluctuation of the *matrix* $\widehat{\Sigma}_n$, not of its individual eigenvalues. To translate matrix-level convergence into eigenvalue-level convergence, we need to know how eigenvalues respond to small perturbations of a symmetric matrix. The cleanest such statement, sufficient for our needs, is a multivariate mean-value theorem for simple eigenvalues.

Lemma 3.5 (First-order perturbation for simple eigenvalues)**.** Let $\boldsymbol{A}$ be a real symmetric $p \times p$ matrix with a simple eigenvalue λ and unit eigenvector $\boldsymbol{u}$. For any real symmetric $\boldsymbol{H}$ with sufficiently small $\|\boldsymbol{H}\|$, the eigenvalue $\lambda(\boldsymbol{A}+\boldsymbol{H})$ of $\boldsymbol{A}+\boldsymbol{H}$ closest to λ satisfies

$$\lambda(\boldsymbol{A}+\boldsymbol{H}) \;=\; \lambda \;+\; \boldsymbol{u}^\top \boldsymbol{H}\,\boldsymbol{u} \;+\; O(\|\boldsymbol{H}\|^2).$$

A self-contained proof requires basic implicit-function-theorem machinery; we give a direct 2×2 verification below, and refer to Abry et al. (2024, Lem. B.10) for the general statement.

Example 3.6 (Verification in the 2×2 case)**.** Take $\boldsymbol{A} = \operatorname{diag}(1,2)$, so the top eigenvalue is $\lambda = 2$ with eigenvector $\boldsymbol{u} = \boldsymbol{e}_2 = (0,1)^\top$. For a symmetric perturbation $\boldsymbol{H} = \begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix}$, the 2×2 eigenvalue formula (6) applied to $\boldsymbol{A} + \boldsymbol{H}$ gives

$$\lambda_+(\boldsymbol{A} + \boldsymbol{H}) \;=\; \frac{(1 + h_{11}) + (2 + h_{22})}{2} \;+\; \frac{1}{2}\sqrt{(1 + h_{11} - 2 - h_{22})^2 + 4h_{12}^2}.$$

For small $\boldsymbol{H}$, Taylor-expanding the square root,

$$\sqrt{(-1 + h_{11} - h_{22})^2 + 4h_{12}^2} \;=\; \sqrt{1 + 2(h_{22} - h_{11}) + O(\|\boldsymbol{H}\|^2)} \;=\; 1 + (h_{22} - h_{11}) + O(\|\boldsymbol{H}\|^2).$$

Substituting back,

$$\lambda_+(\boldsymbol{A} + \boldsymbol{H}) \;=\; \frac{3 + h_{11} + h_{22}}{2} \;+\; \frac{1 + h_{22} - h_{11}}{2} \;+\; O(\|\boldsymbol{H}\|^2) \;=\; 2 \;+\; h_{22} \;+\; O(\|\boldsymbol{H}\|^2).$$

This is exactly Lemma 3.5 with $\boldsymbol{u}^\top \boldsymbol{H} \boldsymbol{u} = \boldsymbol{e}_2^\top \boldsymbol{H} \boldsymbol{e}_2 = h_{22}$.

Top-eigenvalue fluctuations: repeated vs. distinct. With Proposition 3.4 and Lemma 3.5 in hand, the dichotomy between repeated and distinct population eigenvalues is straightforward.

Repeated case: $\Sigma = I_2$. Lemma 3.5 does *not* apply, because both eigenvalues equal 1. Instead, the top-eigenvalue map $\boldsymbol{A} \mapsto \lambda_{\max}(\boldsymbol{A})$ is a continuous function from symmetric matrices to $\mathbb{R}$ (this is the eigenvalue-continuity statement of Proposition 2.19(1)). By the continuous mapping theorem (Proposition 2.6), applied to Proposition 3.4,

$$\sqrt{n}\,(\lambda_{\max}(\widehat{\Sigma}_n) - 1) \;\xrightarrow{d}\; \lambda_{\max}(\boldsymbol{G}),$$

where $\boldsymbol{G} \sim \mathrm{GOE}_2$. The right side is the top eigenvalue of a GOE_2 matrix, which is *not* Gaussian; it has the slightly right-skewed density shaped by repulsion.

Distinct case: $\Sigma = \operatorname{diag}(1,2)$. The population eigenvalues are simple. Lemma 3.5 applies with $\boldsymbol{u} = \boldsymbol{e}_2$, giving

$$\lambda_2(\widehat{\Sigma}_n) - 2 \;=\; (\widehat{\Sigma}_n)_{22} - 2 \;+\; O_p(\|\widehat{\Sigma}_n - \Sigma\|^2).$$

Multiplying by $\sqrt{n}$ and using $(\widehat{\Sigma}_n - \Sigma)_{22} = O_p(n^{-1/2})$ (so the quadratic remainder is $O_p(n^{-1/2})$ and vanishes in the limit),

$$\sqrt{n}\,(\lambda_2(\widehat{\Sigma}_n) - 2) \;=\; \sqrt{n}\,((\widehat{\Sigma}_n)_{22} - 2) \;+\; o_p(1). \tag{12}$$

The entry $(\widehat{\Sigma}_n)_{22} = n^{-1}\sum_i X_{i2}^2$ with $X_{i2} \sim \mathcal{N}(0,2)$ has, by the univariate CLT, $\sqrt{n}\,((\widehat{\Sigma}_n)_{22} - 2) \xrightarrow{d} \mathcal{N}(0,\, \operatorname{Var}(X_{i2}^2))$, and

$$\operatorname{Var}(X_{i2}^2) \;=\; \mathbb{E}[X_{i2}^4] \;-\; 4 \;=\; 12 \;-\; 4 \;=\; 8,$$

using $\mathbb{E}[Z^4] = 3\sigma^4 = 3 \cdot 4 = 12$ for $Z \sim \mathcal{N}(0,2)$. Combining with (12),

$$\sqrt{n}\,(\lambda_2(\widehat{\Sigma}_n) - 2) \;\xrightarrow{d}\; \mathcal{N}(0,\, 8). \tag{13}$$

The limit is now an ordinary Gaussian. An identical argument applied to the smaller eigenvalue gives $\sqrt{n}\,(\lambda_1(\widehat{\Sigma}_n) - 1) \xrightarrow{d} \mathcal{N}(0,2)$, and the two limits are independent because the relevant entries of $\widehat{\Sigma}_n$ are uncorrelated.

Figure 13 confirms this dichotomy with normal Q-Q plots from a Monte Carlo simulation.

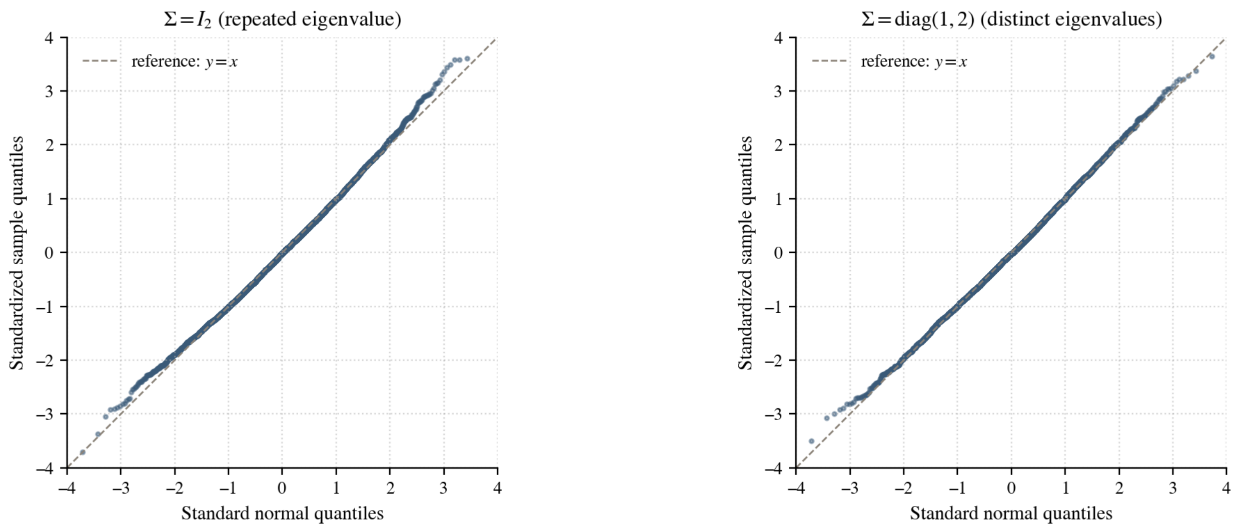

Figure 13: Normal Q-Q plots for the standardized top sample eigenvalue $\sqrt{n}\,(\lambda_{\max}(\widehat{\Sigma}_n) - \lambda_{\max}(\Sigma))$, 5000 Monte Carlo trials at $n = 500$. *Left:* $\Sigma = I_2$ (repeated population eigenvalues). The visible curvature away from the dashed reference line, especially at the right tail, is the signature of the non-Gaussian GOE_2 top-eigenvalue limit. *Right:* $\Sigma = \mathrm{diag}(1, 2)$ (distinct population eigenvalues). The points lie close to the reference line, confirming the Gaussian limit of equation (13). Generated by `notebooks/03_section3_demos.py`.

Eigenvector convergence. A parallel dichotomy applies to eigenvectors. In the distinct case $\Sigma = \mathrm{diag}(1, 2)$, the population eigenvectors are $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$, uniquely determined up to sign. The same first-order perturbation that produced (12) also produces, after a sign convention $\langle \widehat{\boldsymbol{u}}_\ell, \boldsymbol{e}_\ell \rangle \geq 0$,

$$\widehat{\boldsymbol{u}}_\ell \xrightarrow{\text{a.s.}} \boldsymbol{e}_\ell \qquad (\ell = 1, 2).$$

The general statement is in Didier and Pipiras (2012, Lem. B.2).

In the repeated case $\Sigma = I_2$ there is no canonical limiting eigenbasis: every direction in $\mathbb{R}^2$ is a population eigenvector. The sample eigenvectors do not converge, even though the sample matrix itself does, as the following deterministic counterexample makes vivid.

Example 3.7 (Sample eigenvectors can fail to converge under repeated eigenvalues)**.** Define a sequence of 2×2 symmetric matrices

$$\boldsymbol{S}_n = \boldsymbol{O}_n \begin{pmatrix} 1 + \frac{1}{n} & 0 \\ 0 & 1 - \frac{1}{n} \end{pmatrix} \boldsymbol{O}_n^\top, \qquad \boldsymbol{O}_n = \begin{pmatrix} \cos\theta_n & -\sin\theta_n \\ \sin\theta_n & \cos\theta_n \end{pmatrix}, \quad \theta_n = \frac{n\pi}{2}.$$

Then $\boldsymbol{S}_n \to I_2$ in operator norm because $\|\boldsymbol{S}_n - I_2\|_{\mathrm{op}} = 1/n \to 0$. The eigenvectors of $\boldsymbol{S}_n$, however, are the columns of $\boldsymbol{O}_n$; they rotate by $\pi/2$ at every step and never settle. The matrix converges; its eigenvectors do not.

The lesson of Example 3.7 is that eigenvectors require *spectral separation*: without a gap between population eigenvalues, the eigenvectors are not stable under perturbation. This theme returns in Section 4, where only above a critical signal level do sample eigenvectors carry reliable information about the population spike direction.

3.3 High Dimensions: Marchenko-Pastur Law and the Tracy-Widom Edge

The picture of Section 3.2 breaks down qualitatively when p is allowed to grow with n. The basic phenomenon was already visible in Figure 1 of the introduction: with $p = 500$ and $n = 1000$, the sample eigenvalues spread across the interval $[0.086, 2.914]$ even when $\Sigma = I_p$. They no longer converge to $\lambda_i(\Sigma) = 1$. The relevant observable is no longer an individual eigenvalue but the entire eigenvalue distribution, and the natural framework is the *empirical spectral distribution*.

The setup throughout is the *high-dimensional regime*: $p, n \to \infty$ with

$$\frac{p}{n} \longrightarrow y \in (0, \infty).$$

The ratio y is the *aspect ratio* of the data.

Definition 3.8 (Empirical spectral distribution)**.** The *empirical spectral distribution* (ESD) of a $p \times p$ symmetric matrix A with eigenvalues $\lambda_1, \ldots, \lambda_p$ is the probability measure

$$F_A = \frac{1}{p} \sum_{i=1}^{p} \delta_{\lambda_i},$$

placing equal mass $1/p$ at each eigenvalue.

The Marchenko-Pastur law. The fundamental high-dimensional fact is that the ESD of $\widehat{\Sigma}_n$ has a non-trivial limit, even in the null case $\Sigma = I_p$. Figure 14 shows this empirically: the histogram of all $p = 500$ eigenvalues from a single realization closely matches a specific theoretical density on the interval $[(1 - \sqrt{y})^2, (1 + \sqrt{y})^2]$.

The theoretical statement is the Marchenko-Pastur law.

Theorem 3.6 (Marchenko-Pastur law for the null covariance (Marchenko and Pastur, 1967))**.** Let $\boldsymbol{X} \in \mathbb{R}^{p \times n}$ be a Gaussian matrix with iid $\mathcal{N}(0, 1)$ entries, and let $\widehat{\Sigma}_n = (1/n)\, \boldsymbol{X}\boldsymbol{X}^\top$. As $p, n \to \infty$ with $p/n \to y \in (0, \infty)$, the ESD $F_{\widehat{\Sigma}_n}$ converges weakly almost surely to the *Marchenko-Pastur law* MP_y, the probability distribution on $[0, \infty)$ with density

$$f_{\mathrm{MP}_y}(\lambda) = \frac{1}{2\pi y \lambda} \sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}, \qquad \lambda \in [\lambda_-, \lambda_+], \tag{14}$$

where $\lambda_\pm = (1 \pm \sqrt{y})^2$. If $y > 1$, the law has an additional atom at zero of mass $1 - 1/y$.

A complete proof passes through the Stieltjes transform of $F_{\widehat{\Sigma}_n}$ and is beyond the scope of this report; see Yao et al. (2015, Thm. 2.5) and Vershynin (2018, Thm. 3.6). We use the result as a black box.

Three features of the MP law play a role in later sections, and each is visible in Figure 14.

- The support is the interval $[(1 - \sqrt{y})^2, (1 + \sqrt{y})^2]$, of width $4\sqrt{y}$. As $y \to 0$ the support collapses to the point $\lambda = 1$, recovering the low-dimensional consistency of Section 3.2.

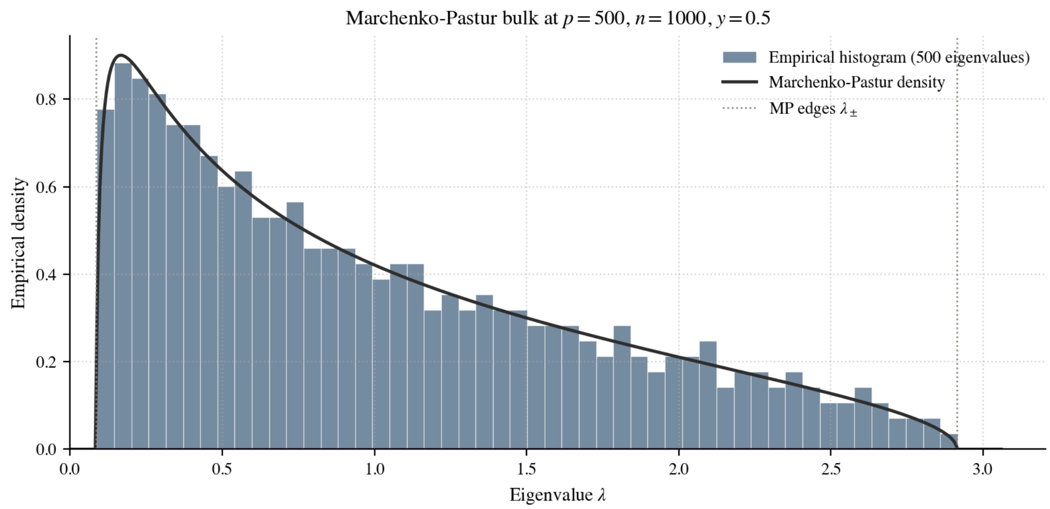

Figure 14: The ESD of $\widehat{\Sigma}_n = (1/n)\,\boldsymbol{X}\boldsymbol{X}^\top$ with iid Gaussian $\boldsymbol{X}$ at $p = 500$, $n = 1000$ (so $y = 1/2$). The histogram of all 500 eigenvalues from one realization closely matches the Marchenko-Pastur density (14) (solid curve). The dotted lines mark the support edges $\lambda_\pm = (1 \pm \sqrt{y})^2$, with $\lambda_- \approx 0.086$ and $\lambda_+ \approx 2.914$. Generated by `notebooks/03_section3_demos.py`.

- Near the upper edge λ_+, the density vanishes like a square root: $f_{\mathrm{MP}_y}(\lambda) \sim C(\lambda_+ - \lambda)^{1/2}$. This square-root edge is the input to the Tracy-Widom heuristic below.
- When $y > 1$ the atom at zero has mass $1 - 1/y$; the matrix $\boldsymbol{X}\boldsymbol{X}^\top$ has rank at most $n < p$, so $p - n$ of its eigenvalues are exactly zero.

The largest sample eigenvalue exits the bulk at the upper edge.

Theorem 3.7 (Convergence of the top eigenvalue (Baik and Silverstein, 2006))**.** Under the hypotheses of Theorem 3.6,

$$\lambda_{\max}(\widehat{\Sigma}_n) \xrightarrow{\text{a.s.}} \lambda_+ \;=\; (1 + \sqrt{y})^2.$$

The Tracy-Widom edge. Theorem 3.7 settles where $\lambda_{\max}(\widehat{\Sigma}_n)$ lives in the limit but says nothing about the fluctuations. Those fluctuations are not Gaussian, and the natural scale is not $\sqrt{n}$ but $p^{-2/3}$. Before stating the precise theorem, we explain the scale via a short heuristic that uses only the square-root edge of the MP density.

Heuristic for the $p^{-2/3}$ edge scale. Near the upper edge,

$$f_{\mathrm{MP}_y}(\lambda) \;\sim\; C\,(\lambda_+ - \lambda)^{1/2} \qquad (\lambda \uparrow \lambda_+).$$

The largest sample eigenvalue is the rightmost of p draws from this density. Heuristically, the mass to the right of $\lambda_{\max}$ should be of order $1/p$:

$$\frac{1}{p} \;\sim\; \int_{\lambda_{\max}}^{\lambda_+} f_{\mathrm{MP}_y}(\lambda)\,d\lambda \;\sim\; C\int_{\lambda_{\max}}^{\lambda_+} (\lambda_+ - \lambda)^{1/2}\,d\lambda \;=\; \tfrac{2C}{3}\,(\lambda_+ - \lambda_{\max})^{3/2}.$$

Solving for the gap,

$$\lambda_+ - \lambda_{\max} \sim p^{-2/3}.$$

The exponent 3/2 in the integrated square-root density inverts to the exponent 2/3 in the fluctuation scale. The same exponent arises in several other random matrix problems; the family of limit laws it generates is the *Tracy-Widom universality class.*

The precise theorem comes with explicit centering and scaling constants from Johnstone (2001).

Theorem 3.8 (Tracy-Widom edge for the null sample covariance matrix (Johnstone, 2001)**.** Under the hypotheses of Theorem 3.6, define

$$\mu_{np} = \frac{(\sqrt{n-1}+\sqrt{p})^2}{n}, \qquad \sigma_{np} = \frac{\sqrt{n-1}+\sqrt{p}}{n}\left(\frac{1}{\sqrt{n-1}}+\frac{1}{\sqrt{p}}\right)^{1/3}. \tag{15}$$

Then

$$\frac{\lambda_{\max}(\widehat{\Sigma}_n)-\mu_{np}}{\sigma_{np}} \xrightarrow{d} \mathrm{TW}_1,$$

where TW_1 is the (real) Tracy-Widom distribution.

The Tracy-Widom TW_1 law was first identified by Tracy and Widom (1996) as the limit of the largest eigenvalue of the GOE; it has approximate mean -1.21 and standard deviation 1.27, and is asymmetric, with a heavier left tail. The centering μ_{np} and scaling σ_{np} converge to $(1+\sqrt{y})^2$ and an $O(p^{-2/3})$ quantity respectively, matching the heuristic above. Figure 15 shows the convergence in simulation.

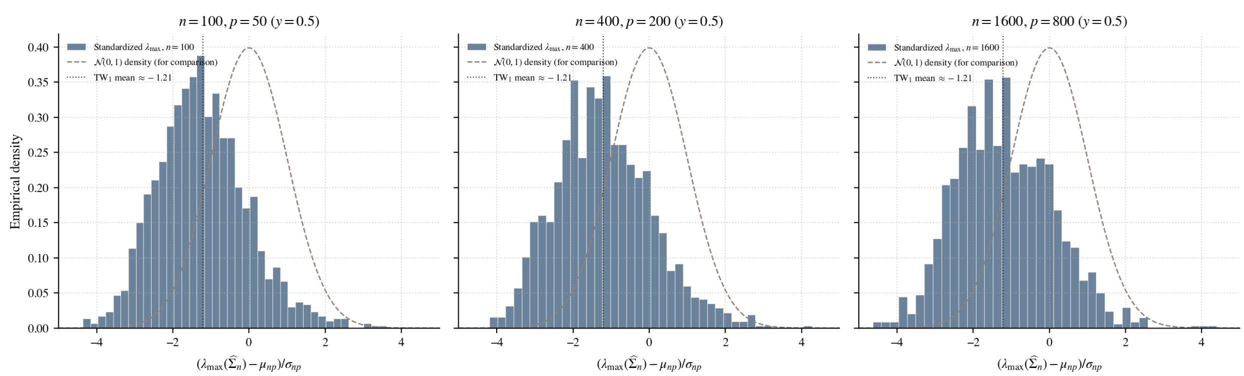

Figure 15: The Tracy-Widom edge in simulation. Each panel shows a histogram of the standardized top sample eigenvalue $(\lambda_{\max}(\widehat{\Sigma}_n) - \mu_{np})/\sigma_{np}$ over $R = 1500$ Monte Carlo trials, with p, n growing at fixed aspect ratio $y = 1/2$. The histograms collapse to a common left-skewed shape that is not Gaussian (compare against the dashed $\mathcal{N}(0,1)$ density); the TW_1 mean ≈ -1.21 is shown for reference. Generated by `notebooks/03_section3_demos.py`.

The picture so far. Combining Sections 3.2 and 3.3, the two regimes look very different.

- *Low dimensions.* Sample eigenvalues are consistent estimators of population eigenvalues. Fluctuations are Gaussian on the $\sqrt{n}$ scale at a simple eigenvalue (via Lemma 3.5) and GOE-shaped at a repeated one (via the continuous mapping theorem applied to Proposition 3.4).
- *High dimensions, null covariance.* The entire empirical spectrum is described by the Marchenko-Pastur bulk on $[\lambda_-, \lambda_+]$, and the top eigenvalue sits at the right edge λ_+ with Tracy-Widom TW_1 fluctuations on the $p^{-2/3}$ scale.

This is the *null* picture: $\Sigma = I_p$ carries no structure. The remaining question is what happens when Σ does carry structure, in particular when a small number of eigenvalues stand above an otherwise flat unit baseline. That setting is the spiked covariance model of Johnstone (2001), and it is the subject of Section 4.

4 Spiked Covariance Matrices and the BBP Transition

[TODO] *Opening paragraph: the Marchenko-Pastur law of Section 3.3 described the null case $\Sigma = I_p$, where all population eigenvalues are equal. In applications the population covariance carries structure: a small number of distinguished directions of high variance riding above a flat noise baseline. The* spiked covariance model *of Johnstone (2001) formalizes this,*

$$\Sigma = \mathrm{diag}(\alpha_1, \dots, \alpha_m, 1, \dots, 1),$$

with m fixed as $p \to \infty$. The central question is when these population spikes are detectable from the sample spectrum, and when the sample eigenvectors carry meaningful information about the population spike directions. The answer is a clean phase transition.

4.1 Johnstone's Spiked Model and the Spike-Location Map

[TODO] *Content to fill in (from v5 + Yao-Zheng-Bai Ch. 11 + Gustavo notes): - Definition: Johnstone's spiked model $\Sigma = \mathrm{diag}(\alpha_1, \dots, \alpha_m, 1, \dots, 1)$ with $H = \delta_1$ for the bulk limiting distribution. - Definition: spike-location map $\Psi(\alpha) = \alpha + \frac{y\alpha}{\alpha-1}$. - Compute $\Psi'(\alpha) = 1 - \frac{y}{(\alpha-1)^2}$. - Lemma: $\Psi'(\alpha) > 0 \iff |\alpha - 1| > \sqrt{y}$; for large spikes $\alpha > 1$ this becomes $\alpha > 1 + \sqrt{y}$. - Verify boundary behavior: $\Psi(1 \pm \sqrt{y}) = (1 \pm \sqrt{y})^2$, so the spike map agrees with the MP edges at the threshold.*

4.2 The BBP Phase Transition for Eigenvalues

[TODO] *Content to fill in: - Theorem (Yao-Zheng-Bai Corollary 11.4 specialized to Johnstone): For a large spike $\alpha_j > 1 + \sqrt{y}$,*

$$\lambda_i(\widehat{\Sigma}_n) \xrightarrow{a.s.} \Psi(\alpha_j), \quad i \in J_j.$$

For $1 < \alpha_j \le 1 + \sqrt{y}$,

$$\lambda_i(\widehat{\Sigma}_n) \xrightarrow{a.s.} (1 + \sqrt{y})^2.$$

Brief sketch (cite Baik and Silverstein (2006) for the proof). - Worked example: $y = 1/2$, so $1 + \sqrt{y} \approx 1.707$ and edge $(1 + \sqrt{y})^2 \approx 2.914$. For $\alpha = 2.5$, $\Psi(2.5) \approx 3.333$. For $\alpha = 1.4$, $\alpha < 1.707$ so the spike is absorbed into the edge. - Simulations (existing figures from v5): Figure: smoothed m=1 simulation, strong vs. weak spike (`smoothed_m1_strong_vs_weak.png`). Figure: raw top eigenvalue distributions (`m1_strong_spike.png`, `m1_weak_spike.png`). Figure (optional): $m = 2$ cases (`m2_two_strong_spikes.png`, `m2_mixed_spikes.png`). Cite Baik et al. (2005); Yao et al. (2015).

4.3 Fluctuations of Spiked Eigenvalues

[TODO] *Content to fill in (from Gustavo's promised sketch + v5): - Statement (fundamental spike, qualitative): for $\alpha > 1 + \sqrt{y}$ in the Gaussian Johnstone model,*

$$\sqrt{n}(\lambda_{\max}(\widehat{\Sigma}_n) - \Psi(\alpha)) \xrightarrow{d} \mathcal{N}(0, \sigma^2_{\alpha,y}).$$

Cite Yao et al. (2015, Thm. 11.11) (fundamental spikes) and Féral and Péché (2009, Thm. 1.6) (cleanest diagonal case). - Heuristic: a separated eigenvalue is a smooth functional of $\widehat{\Sigma}_n$, so a delta-method argument gives $\sqrt{n}$ Gaussian fluctuations (compare with Section 3.2(e)). - Non-fundamental spike: top eigenvalue attaches to the edge, $\lambda_{\max} - \lambda_+ = O_p(p^{-2/3})$, and the fluctuation is expected to be Tracy-Widom type (cite Johnstone (2001)). - Honest note: the critical case $\alpha = 1 + \sqrt{y}$ is not resolved in our current treatment; cite open literature. - Simulation: histogram and Gaussian Q-Q for strong-spike fluctuations (notebook 06).

4.4 Eigenvector Alignment Beyond the Threshold

[TODO] *Content to fill in (from v5 + Yao-Zheng-Bai Thm 11.5/Cor 11.6): - Set-up: in the one-spike model, the population spike direction is $\boldsymbol{e}_1$. Let $\widehat{\boldsymbol{u}}_1$ be the top sample eigenvector. The squared overlap $|\langle \widehat{\boldsymbol{u}}_1, \boldsymbol{e}_1 \rangle|^2 = \cos^2\theta$ removes the sign ambiguity. - Qualitative theorem (Yao-Zheng-Bai Thm. 11.5, Cor. 11.6): strong spike $\Rightarrow$ overlap converges to a positive constant; weak spike $\Rightarrow$ overlap converges to zero. - The one-spike Johnstone overlap formula:*

$$\rho(\alpha, y) = \frac{1 - \dfrac{y}{(\alpha-1)^2}}{1 + \dfrac{y}{\alpha - 1}}.$$

Positive precisely when $\alpha > 1+\sqrt{y}$. - Simulation (figure already in v5/figures/): `eigenvector_overlap_dist` *shows strong-spike overlaps clustered around 0.55–0.65 and weak-spike overlaps near 0. Include and discuss.*

4.5 Sketch of the Connection to Spectral Clustering

[TODO] *Content to fill in (from Gustavo's email + v5 + von Luxburg 2007): - Setup: data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$, kernel similarity $K_{ij} = f(\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2/p)$. - Definition: degree matrix $D = \mathrm{diag}(K\mathbf{1})$ and normalized Laplacian-style kernel $L = nD^{-1/2}KD^{-1/2}$. - Heuristic decomposition: $L =$ noise bulk $+$ low-rank class signal, in direct analogy with $\widehat{\Sigma}_n =$ noise $+$ spike. - Conclusion: class structure is recoverable from L precisely when the signal-to-noise ratio crosses an analogue of the BBP threshold; the informative eigenvectors of L then provide the clustering. Cite von Luxburg (2007) for the algorithmic side and Couillet and Benaych-Georges (2016) for the high-dimensional analysis. - Toy simulation: notebook 08, a two-class Gaussian mixture and the top non-trivial eigenvector of L.*

5 Conclusion and Outlook

[TODO] *Content to fill in: (a) Summary table: the strong vs. weak spike contrast, three rows: eigenvalue limit, eigenvector overlap, fluctuation scale.*

— Spike strength — Eigenvalue limit — Overlap with $\boldsymbol{e}_1$ — Fluctuation — ———————-—————————-——————————————-——————-— — Strong $\alpha > 1+\sqrt{y}$ — $\Psi(\alpha)$ — $\rho(\alpha, y) > 0$ — Gaussian, $\sqrt{n}$ — — Weak $1 < \alpha \leq 1+\sqrt{y}$ — $(1+\sqrt{y})^2$ — 0 — Edge / Tracy-Widom, $p^{-2/3}$ —

(b) One-paragraph synthesis: the BBP phase transition is the cleanest quantitative example of when high-dimensional inference recovers structure and when it does not. It explains, in a controlled toy model, why methods like PCA and spectral clustering succeed only when signal exceeds a threshold determined by the aspect ratio.

(c) Outlook: open questions to flag. - The critical case $\alpha = 1+\sqrt{y}$. The current literature does not yet contain a clean limit theorem; cite open work. - Exact variance $\sigma^2_{\alpha,y}$ in the strong-spike CLT. - Beyond Johnstone's case: general spiked models with non-flat bulk (cite Baik and Silverstein (2006); Benaych-Georges and Nadakuditi (2011)). - Rigorous spectral clustering derivations beyond the toy model; cite Couillet and Benaych-Georges (2016).

(d) Closing sentence: the goal of this thesis was to develop the foundational picture from one end of the dimension spectrum to the other; the open questions above point to where the field is going.

A Auxiliary Results

[TODO] *Content to fill in:*

A.1 Borel-Cantelli Lemmas. - First Borel-Cantelli: if $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$. - Second Borel-Cantelli (independence version): if events are independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$. - Used in Section 2.4 for the convergence-in-probability but not-almost-surely counterexample.

A.2 Slutsky's theorem. - Statement and a one-paragraph proof or reference to Casella and Berger (2002, Thm. 5.5.17).

A.3 Continuous mapping theorem. - Statement: if $X_n \xrightarrow{d} X$ and g is continuous on a set of $\mathbb{P}_X$-probability one, then $g(X_n) \xrightarrow{d} g(X)$.

A.4 Multivariate delta method. - Statement and reference; used in Sections 3.2 and 4.3.

A.5 Characteristic functions and the Lévy continuity theorem. - Definition: $\varphi_X(t) = \mathbb{E}[e^{itX}]$; always exists. - Examples: $\mathcal{N}(0,1)$ gives $e^{-t^2/2}$, Cauchy gives $e^{-|t|}$. - Lévy: $X_n \xrightarrow{d} X \iff \varphi_{X_n}(t) \to \varphi_X(t)$ pointwise, with continuity at 0 sufficient for the converse. - Proof of the classical CLT via $(1+t/n)^n \to e^t$ expansion.

B Simulation Methodology

[TODO] *Content to fill in:*

B.1 General reproducibility conventions. - All notebooks use Python 3, with `numpy`*,* `matplotlib`*, and* `scipy.stats`*. - Random number generation:* `rng = np.random.default_rng(seed)`*, with explicit* `seed` *at the top of every notebook. - Eigendecomposition:* `np.linalg.eigh` *on symmetric inputs; the top eigenpair is taken as* `evals[-1]`*,* `evecs[:, -1]`*. - Plot styling: shared helper module* `notebooks/thesis_style.py` *applies the six-color palette and sans-serif math labels.*

B.2 Per-notebook parameter table. - Notebook 01 (LLN/CLT demos): $n \in \{100, 1000, 10000\}$*, 500 trials. - Notebook 02 (Wishart and repulsion):* $n = 100$*,* $p = 2$*, 5000 trials. - Notebook 03 (fixed-dim fluctuations):* $n = 500$*, 5000 trials, both* $\Sigma = I_2$ *and* $\Sigma = \mathrm{diag}(1, 2)$*. - Notebook 04 (Marchenko-Pastur & TW):* $n = 1000$*,* $y = 1/2$*, 1 trial for bulk density, 1000 trials for top eigenvalue. - Notebook 05 (spiked eigenvalues, BBP):* $y = 1/2$*,* $n \in \{100, 200, 400, 800, 1200\}$*, 200 trials per* n*,* $\alpha \in \{1.4, 2.5\}$*. - Notebook 06 (fluctuation Q-Q):* $n = 1200$*,* $y = 1/2$*, 500 trials,* $\alpha = 2.5$*. - Notebook 07 (eigenvector overlap):* $n = 1200$*,* $y = 1/2$*, 500 trials,* $\alpha \in \{1.4, 2.5\}$*. - Notebook 08 (spectral clustering toy): two-class Gaussian mixture,* $n = 600$ *per class,* $p \in \{50, 200\}$*.*

B.3 Numerical caveats. - For the strong-spike fluctuation Q-Q plot, we standardize empirically (subtract sample mean, divide by sample standard deviation) because the exact variance $\sigma^2_{\alpha,y}$ *requires constants from Yao et al. (2015, Thm. 11.11). - For the weak spike, we do not attempt an exact Tracy-Widom fit because the finite-sample centering and scaling constants are delicate; we report the qualitative shape of the histogram and defer the rigorous fit to future work. - The realized aspect ratio* p/n *differs from the target* y *when* yn *is not integer; figures use the realized ratio in the bulk-edge formula* $(1 + \sqrt{p/n})^2$ *for consistency.*

C Selected Code Listings

[TODO] *Include 4 to 6 of the most informative snippets from the notebooks (not full notebooks). Each listing should be self-contained at the function level, captioned, and referenced from the main text where the corresponding figure appears.*

Suggested listings to include here:

- *C.1 One-spike Johnstone simulation (driving function for Section 4.2). Source:* `notebooks/05_spiked.`
- *C.2 Eigenvector overlap simulation (Section 4.4). Source:* `notebooks/07_eigenvector_overlap.ipy`
- *C.3 Strong-spike fluctuation simulation (Section 4.3). Source:* `notebooks/06_fluctuation_qq_plots`
- *C.4 Marchenko-Pastur bulk simulation (Section 3.3). Source:* `notebooks/04_marchenko_pastur_trac`

Each listing will be wrapped in a `lstlisting` *environment using the* `thesisPython` *style defined in* `style/thesis.sty`*, with a caption and a* `label` *for cross-referencing from the main text.*

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